

The Peering Cliff: A Diagnostic Framework for Internet Transit Policy in the Global South

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Abstract.

Wholesale internet prices remain substantially elevated across the Global South despite a decade of submarine cable investment. I argue that cable capacity is a background enabling condition in most markets studied; the binding constraints on competitive wholesale pricing are gateway governance—who controls access to the landing station—and IXP aggregation scale. A Nash bargaining model formalizes this mechanism, deriving a price-discontinuity threshold, the “Peering Cliff,” at a traffic asymmetry ratio of approximately 2:1, above which settlement-free peering is not individually rational for Tier-1 carriers, and yields an investment-sequencing diagnostic applicable at the pre-project stage without proprietary data. Three sources provide mechanism-consistent evidence. The 2Africa cable landing—the world’s largest submarine system—produced no average price response across ten Sub-Saharan African markets. Countries with carrier-neutral IXPs at landing show an implied price differential of approximately 32 percentage points relative to those without; IXP-CN countries were better-governed and more developed on every observable pre-treatment dimension, so this estimate is an upper bound on the IXP effect rather than a causal estimate. Before-after comparisons for the primary target population are inconclusive once a data artifact is removed; the 2Africa differential is the relevant evidence for that population. A cross-country IXP participant threshold regression identifies a structural break at approximately 30 members, consistent with the model’s aggregation prediction. None of these sources meets the bar for causal identification. The framework’s scope conditions—competitive gateway access, sufficient market scale, and market-responsive carrier pricing—follow from the model’s own structure and indicate where the standard instrument is likely to fail before funds are committed.

JEL Classification: L96, L14, L51, O19, O33

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Summary

Wholesale internet prices in the Global South remain far above developed-country benchmarks. The median developing-country market pays \$1.06 per Mbps for IP transit; London pays \$0.17. This gap persists despite substantial submarine cable investment over the past decade. I ask why, and offer a practical answer for development finance practitioners designing digital infrastructure projects.

The central argument is that cable capacity is a background enabling condition in most markets; the binding constraints on competitive wholesale pricing are gateway governance and IXP aggregation scale. A cable landing under single-operator control does not alter the bargaining position of domestic networks regardless of the capacity it delivers. A carrier-neutral Internet Exchange Point (IXP) at sufficient scale does, by aggregating domestic traffic to the threshold where Tier-1 carriers find settlement-free exchange individually rational—but only where competitive access to the landing station already exists.

A Nash bargaining model formalizes this. It derives a price-discontinuity threshold, the “Peering Cliff,” at a traffic asymmetry ratio of approximately 2:1, above which settlement-free peering is not individually rational for Tier-1 carriers. The model yields four propositions: the threshold exists and is increasing in traffic volume; aggregation across a carrier-neutral IXP can move a market from a locked-in pricing regime to a competitive one; mandated bill-and-keep cannot achieve this because it requires a carrier to accept below-cost terms; and the mechanism operates if and only if three conditions hold— competitive gateway access, sufficient market scale, and market-responsive carrier pricing. These scope conditions are predictions of the model, not post hoc empirical retreats.

Evidence consistent with the mechanism is drawn from three independent sources. The 2Africa cable landing—the world’s largest submarine system—produced no average wholesale price response across ten Sub-Saharan African landing markets. Countries with carrier-neutral IXPs at landing show an implied price differential of approximately 32 percentage points relative to those without; because IXP-CN countries were better-governed and more developed on every

observable pre-treatment dimension, this estimate is an upper bound on the IXP effect rather than a causal estimate. Before-after price comparisons for Sub-Saharan Africa and Central Asia—the framework’s primary target population—are inconclusive once an anomalous data artifact is removed; the 2Africa differential is the mechanism-specific evidence most directly relevant to that population. A cross-country IXP participant threshold regression identifies a structural break at approximately 30 participants, below which markets pay substantially higher prices at equivalent infrastructure levels. Adding regional fixed effects raises cross-country explanatory power tenfold, confirming that institutional and structural factors dominate physical infrastructure as price determinants. None of these sources meets the bar for causal identification; the pre-specified MOZIX prospective test (Section 5.1; Supplementary Materials, Section S13) is a step toward stronger evidence.

The model’s primary application domain is competitive-gateway African markets and structural analogs. It does not apply where state ownership severs market-responsive pricing—as in Viet Nam and Pakistan, where cable capacity additions produced no price response—or where markets are too thin for national aggregation to reach minimum viable scale. A diagnostic that can identify these failure cases before funds are committed is the operational contribution I make.

1. Introduction

In 2023, the 2Africa cable—the world’s largest subsea system by capacity—landed at Maputo, Mozambique. Two years later, wholesale internet prices in Mozambique have not moved. Maputo still pays \$4.89 per Mbps for IP transit. All three landing stations remain under the control of the state-owned incumbent, Telecomunicações de Moçambique.²

This is not a puzzle about infrastructure. The fiber is there. The puzzle is about who controls the point where that fiber meets the domestic network, and what happens to prices when the answer is: one company, with no competitive pressure to share the gains.

The structural bottleneck is not cable capacity but access to the landing station. A carrier-neutral Internet Exchange Point (IXP) does what a cable cannot: it aggregates domestic network traffic to the threshold where Tier-1 carriers find settlement-free exchange worthwhile, shifting the bargaining position of smaller networks without requiring regulatory intervention. The investment-sequencing error this paper is designed to diagnose is financing cable capacity without ensuring competitive access to the landing station and carrier-neutral aggregation infrastructure. Additional bandwidth that leaves gateway governance unchanged delivers no price benefit, as the 2Africa evidence makes plain. The welfare stakes are substantial: the Global South median price is \$1.06/Mbps while London pays \$0.17/Mbps; Luanda pays \$16.92/Mbps—differentials that have persisted through a decade of infrastructure investment.³ Where retail markets are competitively structured, wholesale price compression passes through to end-users; the size of that benefit depends on retail market conditions that vary across the paper’s target markets and are not estimated here (Section 6.3).

That structural factor is bargaining power at the interconnection point. Networks exchange traffic either by paying transit—a per-Mbps fee for global routing access—or through settlement-free peering, in which two networks exchange traffic at no charge. Peering, however, is conditioned on

²TeleGeography IP Transit Pricing Service, Maputo city-level observations, Q1 2023–Q3 2025. A single repricing event in Q2 2024 established a new floor at \$4.89/Mbps; no further price compression followed over six consecutive quarters. See Figure 8 and Section 5.1.

³The \$1.06/Mbps figure is the median for the 52-country developing-market subsample (non-high-income, non-OECD-region); the full 97-country sample median including high-income countries is \$0.60/Mbps (Table 6).

traffic balance. When a developing-country ISP imports far more content than it exports—because its digital economy consumes global content but produces little of it—the resulting traffic asymmetry ratio is too high for a Tier-1 carrier to accept peering without payment. The smaller network is stuck paying transit at prices that can run 10 to 40 times the developed-market benchmark, regardless of how much cable capacity lands at its shore.

An IXP breaks this trap—but only under the right conditions. By pooling traffic across multiple smaller networks, a carrier-neutral IXP raises the combined volume that any single carrier sees, potentially crossing the asymmetry threshold at which peering becomes rational. Critically, this mechanism depends on carrier neutrality: an IXP co-located with or governance-captured by the incumbent carrier does not provide independent aggregation. Sixty-nine percent of low-income countries had at least one IXP as of 2021 [[World Bank, 2021](#)]—yet prices remain elevated. IXP presence and IXP effectiveness are not the same thing.

This paper is a diagnostic framework, not a causal study. A Nash bargaining model formalizes the mechanism and yields four propositions: the Peering Cliff threshold exists and is proportional to observable traffic volumes; aggregating networks across a carrier-neutral IXP can move an aggregated entity from a locked-in pricing regime to a competitive one; aggregation is incentive-compatible for Tier-1 carriers in ways that mandated bill-and-keep is not; and—the proposition that makes the framework generalizable—the mechanism operates if and only if three conditions derived from the model’s own structure hold (Proposition 4). What the framework can do is indicate which constraint is most likely binding—a distinction that is consequential in development finance, where a project financing cable investment in a market where landing-station control is the operative problem fails because it misdiagnosed the constraint, not because it ran bad regressions. The evidence is mechanism-consistent across multiple independent sources and does not meet the bar for causal identification; Section 5 discusses the evidential standard this implies. A pre-specified prospective test (Section 5.1; Supplementary Materials, Section S13) is underway at the Mozambique IXP, with a four-quarter evaluation window closing in late 2026.

Where the scope conditions hold, the evidence is consistent with the mechanism; where

they fail, the evidence is consistent with their failure. A framework whose scope restrictions are empirically testable—and whose null predictions hold where the model requires them—is more useful to a task team than one that reports only favorable cases.

Contributions.

1. *Theoretical*: A Nash bargaining model derives the Peering Cliff threshold from first principles (Supplementary Appendix A), yielding four propositions: threshold existence and monotonicity; aggregation as an escape mechanism; the incentive incompatibility of mandated peering; and a formal derivation of the scope conditions under which the mechanism operates (Proposition 4). The 2:1 calibration is consistent with documented carrier peering policies [[Norton, 2012](#), [Greenstein, 2015](#)].
2. *Empirical*: Mechanism-consistent evidence from the 2Africa differential treatment effect (a large, exogenous capacity shock that produced no price response absent carrier-neutral IXPs, and an implied 32 percentage-point differential where they were present); an IXP participant threshold regression identifying a structural break at approximately 30 members; and cross-country regressions indicating that institutional factors dominate physical infrastructure as price determinants. The transit-share zone typology provides internal-coherence corroboration; SOE case evidence from Viet Nam and Pakistan confirms the scope condition under which the mechanism does not apply.
3. *Operational*: An investment-sequencing diagnostic mapping observable market structure—gateway ownership, IXP governance and scale, traffic asymmetry range—to three regimes with distinct intervention pathways, applicable without proprietary data or econometric estimation.

2. Literature Review

2.1. Bargaining Theory and the Traffic Ratio Debate

The economics of internet interconnection rests on bargaining between asymmetric parties. The foundational results establish that large networks have strategic incentives to degrade interconnection

quality with smaller rivals [Laffont et al., 1998], that bill-and-keep favors larger networks when traffic is imbalanced [DeGraba, 2000], and that equilibrium terms shift from settlement-free to paid arrangements as asymmetry grows [González, 2000]. Foros and Hansen [2001] extended this framework to backbone market competition, showing that transaction costs and routing costs interact to determine equilibrium interconnection terms—a result bearing on the thin-market setting analyzed here. Gans and King [2001] showed that bill-and-keep can soften network competition under symmetric networks, but that this result breaks down under traffic asymmetry, motivating the focus on asymmetry as the operative constraint.

Industry practice has converged on traffic ratio thresholds between 2:1 and 5:1 [Norton, 2012]. Zarchy et al. [2018] derive Nash-based transfer payments under symmetric outside options, finding that efficient peering prices should reflect cost differentials rather than traffic ratios. That result does not extend to the Global South context, where transit cost asymmetries of $20\times$ between counterparties restore traffic ratios as determinants of equilibrium transfer payments—once asymmetric outside options are restored, Proposition 1 shows formally that the Zarchy result reverses. Greenstein [2015] documents the Level 3–Comcast dispute as a real-world instance of the threshold mechanism, though the regulated institutional context limits its direct analogy to the developing-country setting.

2.2. Coordination Failures in Network Formation

Lodhi [2014] demonstrates through game theory and agent-based simulation that transit providers gravitate toward suboptimal peering arrangements through myopic, uncoordinated decision-making, and that initial conditions determine outcomes: identical traffic conditions can sustain either efficient or inefficient equilibria depending on history. Once providers reach a suboptimal equilibrium, unilateral deviation reduces utility even when mutual deviation would benefit all parties. Countries entering the internet ecosystem with severe traffic asymmetries may stabilize at bad equilibria from which incremental interventions cannot escape—the theoretical basis for coordinated simultaneous intervention across regulatory, physical, and governance dimensions in markets with $R \geq 5$ (Section 5.4).

2.3. Infrastructure, Aggregation, and Internet Pricing

[Cariolle et al. \[2024\]](#) find that submarine cable landings reduce transit prices, but the effect varies substantially across markets. Studies of IXP deployment in Africa document significant savings: KIXP reduced local latency from 200–600ms to under 30ms and generated estimated annual savings of \$1.4 million [[Kende, 2020b](#)]; Nigeria’s IXPN (Internet Exchange Point Nigeria) reduces costs by approximately \$40 million annually [[Kende, 2020a](#)]. The World Development Report 2021 documents that “local bits are cheaper than exported bits,” estimating that Latin America spent approximately \$2 billion annually on international bandwidth reducible by one-third through greater IXP utilization [[Comini et al., 2021](#), [World Bank, 2021](#)].⁴ Research on IXP governance suggests organizational form matters: European IXPs typically operate as non-profit cooperatives; US IXPs more often commercially [[Chatzis et al., 2013](#)]. In Latin America, successful deployment has relied on not-for-profit ISP associations [[Galperin and Fernández-Ardèvol, 2023](#)]. The cooperative model reduces mistrust—a barrier salient in markets where incumbent resistance is common. IXPs improve wholesale prices not primarily through technical efficiency but by strengthening the small network’s outside option: credible threat to route via alternative peers shifts its disagreement point and the resulting Nash equilibrium price.

How much demand responds to wholesale price reductions matters for the welfare magnitude of IXP investment. [Orujov and Jafarov \[2025\]](#) estimate broadband price elasticity in a panel of 33 European countries (2010–2024), finding substantial heterogeneity: Eastern Partnership countries show elasticities around -0.61 , while EU markets average -0.12 .⁵ Applied to the Global South context, high latent elasticities in developing markets imply that wholesale price reductions—if passed through to retail—would generate proportionally larger demand responses than the European estimates suggest, amplifying the welfare case for aggregation investment. This demand-response channel operates independently of the bargaining mechanism this paper formalizes.

⁴This accounting-identity estimate uses 2019 bandwidth cost data; given bandwidth volume growth since 2019, the current figure is likely substantially higher.

⁵[Orujov and Jafarov \[2025\]](#) is available as arXiv preprint 2512.21356 and has not yet completed peer review; the elasticity estimates are treated here as indicative rather than definitive.

The welfare stakes of wholesale price compression extend beyond ISP margins. [Hjort and Poulsen \[2019\]](#) find that terrestrial fiber rollout in Africa increased employment by approximately 9 percentage points in connected areas; [Bahia et al. \[2020\]](#) document comparable effects from mobile broadband expansion. The upstream wholesale pricing mechanism this paper analyzes is a precondition for the access improvements those papers document: the last-mile connectivity gains [Hjort and Poulsen \[2019\]](#) identify require competitive wholesale pricing at the interconnection layer this framework targets. A framework for identifying and correcting the binding constraints on that mechanism is therefore a complement to, not a substitute for, the infrastructure-investment literature.

2.4. State-Owned Enterprises and Soft Budget Constraints

[Kornai \[2014\]](#) shows that enterprises expecting state bailouts face weakened cost-minimization incentives—the soft budget constraint. Applied to international gateway pricing, this predicts that SOE-controlled gateways will not respond to competitive entry or capacity investment in the way the bargaining model implies: pricing decisions are administrative rather than market-responsive, so the mechanism that translates aggregation infrastructure into competitive terms is severed. [Maskin and Xu \[2001\]](#) establish that competition—not ownership transfer—is the primary mechanism for hardening budget constraints. Partial privatization without competitive entry at the relevant market layer (as in Pakistan, Section 5.3) therefore predicts unchanged wholesale pricing behavior. Table 4 (Section 6) distinguishes soft-budget-constraint markets from security-architecture markets on the basis of observable indicators; the required reform pathways differ substantially between the two and are discussed in Section 6.1.

The challenge of sequencing regulatory reform ahead of infrastructure investment has been documented in the African telecom context. [Calandro et al. \[2012\]](#) show that South Africa’s early failure to separate Telkom’s network and retail operations delayed effective broadband competition by nearly a decade; [Gillwald et al. \[2013\]](#) document comparable sequencing failures across a cross-section of African markets, finding that infrastructure investment without regulatory reform of access conditions systematically failed to reduce retail prices. The soft-budget-constraint mechanism

this paper formalizes is not a theoretical novelty: the African empirical record is consistent with the prediction that ownership without competitive discipline at the relevant market layer leaves pricing behavior unchanged.

3. Data and Calibration

The model is calibrated using TeleGeography’s *Global Internet Geography* and *IP Transit Pricing Service*; IXP presence is from PeeringDB.⁶ An IXP is classified as *carrier-neutral* if it satisfies two conditions jointly: (i) physical co-location at a facility not owned or exclusively operated by the dominant domestic carrier, and (ii) open membership on non-discriminatory terms. The model’s Condition (C1) requires an active switching fabric presenting aggregated traffic directly to the Tier-1 carrier—co-location alone does not establish the bilateral peering sessions that constitute direct access.⁷ Traffic flows between developed and developing regions are highly asymmetric (ratios exceeding 10:1), and volume discounts create economies of scale favoring aggregation. Table 1 maps theoretical concepts to empirical indicators.

Data and Sample. The TeleGeography IP Transit Pricing Service is self-reported: carriers with the highest prices may be least likely to report, suggesting documented differentials are conservative. City-level prices aggregated to country level obscure within-country variation. Proprietary data limit independent replication; I provide synthetic data matching summary statistics (Supplementary Appendix C) and a publicly-sourced validation panel.

Data quality and outlier handling. Observations exceeding the regional median by more than three standard deviations are screened using criteria documented in Supplementary Materials, Table S1; three of seven flagged observations are excluded, including Cameroon’s anomalous

⁶Terminology: *Settlement-free peering* = no payment flows; *Transit* = purchasing global routing access per Mbps; *Traffic ratio* (R) = inbound/outbound ratio. The “Peering Cliff” is the threshold ($R \approx 2$) where settlement-free peering becomes unavailable. The zone labels—*Peering Zone*, *Aggregation Zone*, *Locked-in Zone*—refer to traffic asymmetry ranges; a country classified as “Peering Zone” by its transit share may achieve that share through SOE vertical integration rather than competitive peering (Supplementary Materials, Table S11).

⁷IXP misclassification is a real risk: a nominally present IXP that is governance-captured by the incumbent would be coded “IXP present” but fails the carrier-neutrality criterion. The direction of bias is unambiguous—misclassified captured IXPs attenuate the estimated IXP effect toward zero—so effect estimates should be interpreted as conservative lower bounds. The MOZIX case illustrates this: MOZIX is coded “IXP present” throughout the panel despite a governance change in November 2024, which is why it is analyzed separately as a within-country test (Section 5.1) rather than an IXP-formation event.

Table 1: Variable Definition: Theoretical Concepts to Empirical Proxies

Concept	Indicator	Construction	Limitation
Bargaining power	Transit share ^a	% traffic via paid transit vs. peering	Outcome, not input
Outside option (BATNA)	IXP presence	Binary: ≥ 1 IXP in country	Doesn't capture IXP quality or carrier-neutrality (see footnote above)
Cost structure	Int'l capacity	Total lit Gbps (submarine + terrestrial)	Wholesale, not unit cost
Market structure	SOE share	% gateway capacity controlled by SOEs	Classification judgment
Traffic asymmetry (R)	Inferred from transit share	$<40\% \rightarrow R < 2$; $40-70\% \rightarrow R \in [2, 5]$; $>70\% \rightarrow R \geq 5$	Indirect mapping
Coastal access	Cable landing ^b	Submarine cable or >10 Gbps backhaul within 500km	Threshold is approximate

Sources: TeleGeography (prices, capacity, transit share); PeeringDB (IXP presence); ITU and regulatory filings (SOE classification). ^a*Circularity note:* Transit share is an *outcome* of the bargaining process, not an exogenous input; used here as a diagnostic indicator of revealed bargaining position, not a causal variable. ^b*Threshold basis:* The 500km threshold reflects engineering estimates for latency-competitive terrestrial backhaul (<5 ms one-way latency at fibre speeds). Alternative thresholds (200km, 1000km) do not materially affect results.

pre-IXP price of \$73.36/Mbps. The direction of any residual measurement error is conservative: retained outliers bias price differences toward zero.

The regression sample covers 97 of 169 economies for which infrastructure data are available. The missing 72 are heavily weighted toward small island states, fragile states, and low-income countries—the markets the analysis most aims to inform. Selection bias likely runs upward: covered markets have more developed digital ecosystems and stronger institutions, so estimated associations should be read as upper bounds on the expected effect in the most fragile markets. Sample attrition across all analyses is documented in Supplementary Materials, Table S1; no single analysis uses all 169 economies.

4. The Bargaining Model

The model generates testable predictions (threshold existence, comparative statics, policy ranking, and scope conditions), tested against cross-country price patterns, a staggered IXP event study, and the 2Africa capacity shock. I distinguish **proposition-level results**, which are robust to parameterization, from **calibration-specific results**, which are not. A full theory-to-evidence mapping is in Supplementary Materials, Table S0.

Four propositions follow from the model. The first three establish that the threshold exists, that aggregation can cross it, and that mandated peering cannot. The fourth—Proposition 4—derives the conditions under which the bargaining mechanism operates at all; where any condition fails, the model predicts no price response to aggregation, and the empirical evidence is examined against that prediction directly.

A supporting lemma (Supplementary Appendix A.3c) connects these results to the gateway governance problem: the threshold R^* is simultaneously the asymmetry above which j will not offer settlement-free peering *and* the asymmetry above which j has no independent incentive to bypass an incumbent gateway. Gateway market power and traffic asymmetry are therefore self-reinforcing—both are most severe in the high- R markets the analysis addresses, and both are disciplined by the same instrument: carrier-neutral aggregation that raises j 's combined bypass value.

The 2:1 threshold is calibrated to documented carrier peering policies and is consistent with the Level 3–Comcast dispute [Greenstein, 2015], in which a traffic ratio already near 2:1 rose to approximately 5:1 after Level 3 contracted to carry Netflix traffic, triggering contested interconnection. This high-income case illustrates the mechanism but should not be read as an out-of-sample test; the institutional context—FCC oversight, symmetrically-resourced parties, regulated interconnection—differs substantially from the developing-country setting. For planning purposes the threshold should be treated as a range of 1.5:1–3.5:1; policy sequencing recommendations are invariant across this range (Supplementary Materials, Table S5b).

4.1. Model Setup

Networks exchange traffic through three arrangements. *Transit*: a smaller network pays a larger one for global routing access—the defining wholesale cost for most developing-country ISPs. *Settlement-free peering*: two networks exchange traffic at no charge, conditioned on traffic balance. *IXP-mediated exchange*: a carrier-neutral facility through which any connected network can peer with any other via a single physical connection, substituting for local but not international transit. Without a local exchange point, traffic between two ISPs in the same city may route through hubs thousands of kilometers away—KIXP reduced this “tromboning” latency from 200–600ms to under 30ms when it activated [Internet Society and Analysys Mason, 2012].⁸

I model interconnection as a non-cooperative bargaining process where networks compare Direct Negotiation (subject to the asymmetry ratio R) against Indirect Routing (via a Hub). Bilateral bargaining reflects institutional realities in developing markets: weak regulatory oversight, absent standardized contracts, limited price transparency, and asymmetric information about traffic volumes. These conditions make unmediated negotiation the default interconnection mode. Figure 1

⁸The cost-function mechanics are straightforward. Before IXP establishment, an ISP’s total transit cost for handling traffic volume Q is $C = c_T \cdot Q$, where all traffic—including domestic—is routed through international transit at unit cost c_T . After IXP establishment, total traffic bifurcates into domestic (Q_d , exchanged at the IXP) and international (Q_i), and the cost function becomes $C = F_{\text{IXP}} + c_T \cdot Q_i$, where F_{IXP} is the fixed port fee. As long as $c_T \cdot Q_d > F_{\text{IXP}}$ —a condition satisfied in virtually all markets with non-trivial domestic traffic—the ISP’s marginal cost of serving domestic traffic falls to near zero. This cost-function shift is the micro-foundation for the price declines documented in the event study (Section 5.2); the Nash bargaining model formalizes the conditions under which this shift is *achievable* given the strategic environment.

illustrates the mechanism. International bandwidth capacity enters through the Tier-1 carrier’s cost parameter $c_T^j(K)$, which falls with available capacity K up to a saturation point $\bar{K}$ (Supplementary Appendix A, Eq. A.1). Beyond $\bar{K}$, additional capacity does not further reduce the carrier’s costs. In the capacity-abundant regime ($K \geq \bar{K}$), the Peering Cliff threshold R^* depends on traffic asymmetry and IXP aggregation, not on capacity; this is the model as presented below. The 2Africa evidence (Section 5.1) indicates that Sub-Saharan African landing markets are in this regime. The capacity control regressions (Supplementary Materials, Section S7b) confirm that the IXP threshold effect survives when $\ln(K_{ct})$ is controlled for.

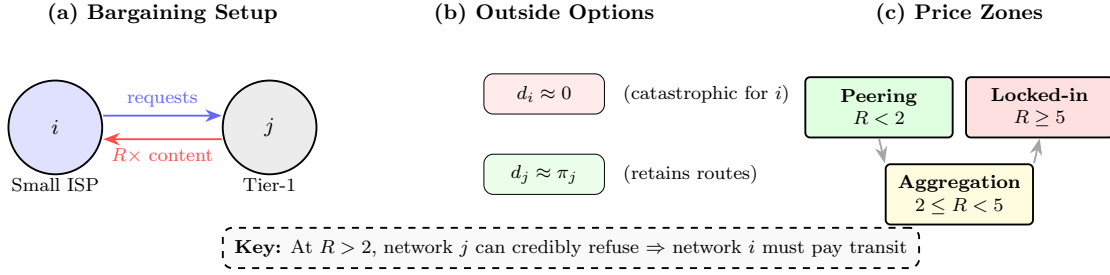


Figure 1: Nash Bargaining Framework for Interconnection. (a) Two networks exchange traffic with asymmetry ratio R . (b) Outside options differ: disconnection is catastrophic for the small network but costless for the large carrier. (c) The resulting price zones depend on R ; the threshold $R^* \approx 2$ marks where bargaining leverage shifts.

4.2. The Peering Cliff: Where Bargaining Power Shifts

Europe and Asia ($R \approx 1.0$) achieve stable peering. Latin America ($R \approx 4.5$ with North America) sits in a transition zone. The Europe-Africa corridor illustrates asymmetry persistence: even after recent IXP growth, $R \approx 10.2$ in 2025, projected to fall only to 8.7 by 2030—still well above the peering threshold.⁹

The 2:1 threshold represents the point where the large network’s outside option approaches its connected profit, eliminating its incentive to offer concessions. I use 5:1 as a conservative *lock-in threshold*: at $R = 5$, the escape-velocity condition requires an IXP with approximately 70–90 active

⁹Derived from TeleGeography IP Traffic Statistics (2025), aggregate inbound-to-outbound ratios for the Europe-to-Africa corridor. The 2030 projection applies estimated growth rate differentials between African outbound and inbound traffic (author’s calculation); both figures remain well above any plausible peering threshold.

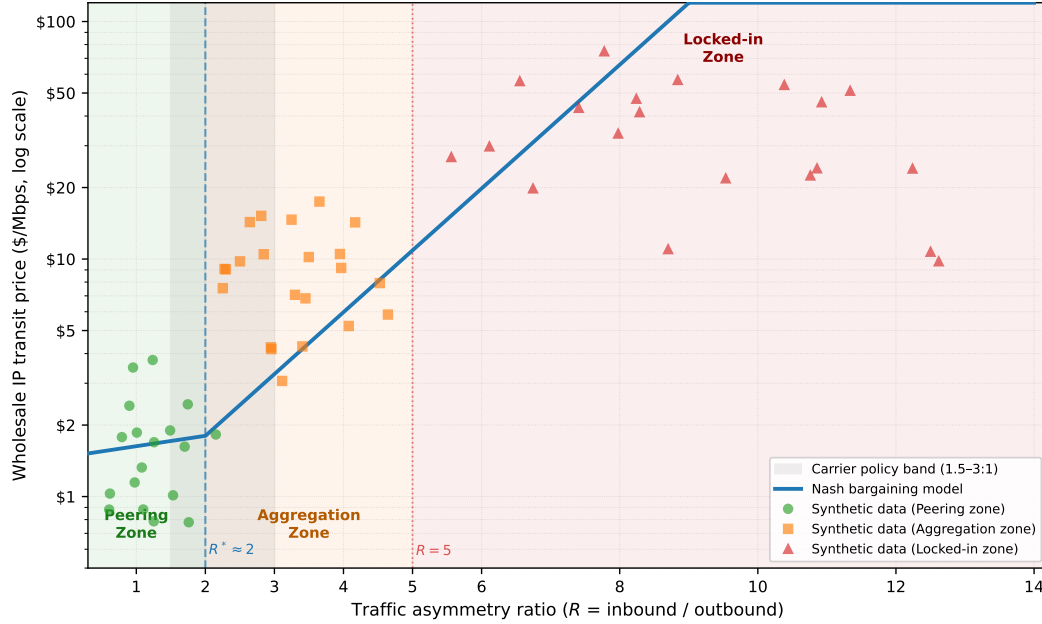


Figure 2: The Peering Cliff: Traffic Asymmetry vs. Wholesale Price. *Horizontal axis*: Traffic asymmetry ratio (R). *Vertical axis*: Wholesale IP transit price (\$/Mbps, log scale). The curve is a theoretical prediction from the Nash bargaining model (Supplementary Appendix A), not a fitted spline. Scatter points are synthetic data calibrated to match the distributional properties of TeleGeography country-level prices; they illustrate the qualitative price-asymmetry relationship and are not independent observations (see Author’s Note on reproducibility). The shaded band (1.5–3:1) represents empirical variation in carrier policies. Policy zones: *Peering Zone* ($R < 2$), *Aggregation Zone* ($2 \leq R < 5$), *Locked-in Zone* ($R \geq 5$). The $R = 2$ boundary is model-derived (Proposition 1); the $R = 5$ boundary is an operational judgment based on $N^*(R)$ scaling, not a model result—see text.

members under the baseline parameterization, a scale not yet achieved in any Sub-Saharan African or Central Asian market. The Locked-in Zone therefore marks the region where aggregation must be combined with structural reforms rather than pursued in isolation.¹⁰

Deriving the Threshold. Network j accepts settlement-free peering if and only if the routing value of i 's return traffic covers the fixed cost of the peering link: $c_T^j \cdot Q_{ij} \geq F$. Substituting $Q_{ij} = Q_{ji}/R$ yields the Peering Cliff threshold (derived in Supplementary Appendix A):

$$R^* = \frac{c_T^j \cdot Q_{ji}}{F} \quad (1)$$

When R exceeds R^* , j demands a positive transfer payment $t^* > 0$. The calibration is illustrative, chosen to be consistent with documented carrier benchmarks (AT&T 2.0:1, Verizon 1.8:1, Lumen 1.5:1) and should be read as a model prediction grounded in carrier policy evidence, not a structural parameter estimate.¹¹ Empirical estimates range from $\hat{R} \approx 2.2$ in the competitive cross-section to $\hat{R} \approx 3.4$ in the full panel, reflecting forward-looking trajectory-risk margins applied by carriers and the entry of SOE-dominated markets via administrative rather than competitive pricing. Policy sequencing recommendations are invariant across $R^* \in [2.0, 3.5]$; full decomposition is in Supplementary Materials, Section S5e. The policy ranking—aggregation over mandated peering—is also invariant to the bargaining power parameter $\lambda \in (0, 1)$ (Supplementary Appendix A.2b).

Internal Consistency Check: Transit-Share Threshold. A structural limitation governs both exercises that follow and should be stated before the results: transit share is an *outcome* of the bargaining process being modeled, not an exogenous input. Mapping an outcome variable back to the structural parameter R to assess a threshold effect borders on tautology—the calibration and the data share the same informational basis. This circularity is not resolved by the scaling sensitivity exercise ($b \in \{3, 4, 5\}$, Supplementary Materials, Section S5c), which confirms that zone classifications are stable across scaling assumptions but does not address the underlying endogeneity.

¹⁰“Peering Cliff” is a stylized label; the underlying Nash bargaining solution is a smooth function of R . The “cliff” refers to the region where $t^*(R)$ rises steeply as R crosses R^* , not to a mathematically discontinuous price jump (Supplementary Appendix A, Figure A1).

¹¹Developing-country ISPs face this threshold at lower absolute traffic volumes because thin digital ecosystems reduce Q_{ji} , tightening R^* ; the welfare cost is amplified by c_T^j values 10–40 $\times$ above developed-market benchmarks.

Both exercises below are therefore internal consistency checks—validating that the descriptive pattern of the data is coherent with the model’s structure—rather than independent tests of the threshold. The transit-share zones should be read as identifying *locked-in versus peered regimes* in the data, not as confirming the precise theoretical $R^* = 2$. The supply-side IXP participant threshold regression (Section 5.2) is the operationally stronger evidence; it uses a variable—member network count—that is not determined by the bargaining outcome the model predicts.

With that framing established: because R is not directly observed, I use transit share as a proxy under the illustrative parameterization $R \approx 1 + 4 \times \text{transit share}$.¹² A prespecified Mann-Whitney test at transit share = 0.25—the theoretically predicted location—finds that countries above this cutoff exhibit median prices approximately 180% higher ($p < 0.01$). An exploratory Hansen grid search locates a breakpoint at $\hat{R} = 2.20$, directionally consistent with the theoretical prediction, but the bootstrap p -value of 0.75 means the threshold location is imprecisely identified.¹³ Both results are consistent with the existence of distinct locked-in and peered pricing regimes; neither constitutes an independent test of the theoretical threshold value.

Mapping Theory to Observables. Transit share is an outcome of the bargaining process (Table 1, note a) and should be read as a diagnostic proxy for R , not as a structural input. With that caveat: “Locked-in Zone” (>70% transit) corresponds to $R \geq 5$; “Aggregation Zone” (40–70%) to $2 \leq R < 5$; “Peering Zone” (<40%) to $R < 2$. SOE-dominated markets can exhibit low transit share via state-controlled backbone routing, producing Peering Zone classification without competitive bargaining. Zone classifications should be validated against structural indicators (landing station ownership, IXP governance) before use; the IXP participant threshold regression (Section 5.2) is the more reliable primary diagnostic.

¹²Derivation, boundary-condition calibration, and sensitivity to alternative scaling parameters $b \in \{3, 4, 5\}$ are in Supplementary Materials, Section S5c. Policy zone classifications are invariant to the choice of b .

¹³Hansen threshold regression: grid search over candidate breakpoints from the 10th to 90th percentile in steps of 0.05; 500-iteration bootstrap (seed: 20260115). The sup-Wald statistic plot is in Supplementary Materials, Section S5b.

4.3. Hub-and-Spoke Efficiency and Thin Markets

A small network that cannot achieve settlement-free peering bilaterally may achieve it by routing through a regional exchange point where aggregated traffic crosses the 2:1 threshold—the hub-and-spoke corollary to the aggregation mechanism. NAPAfrica (Johannesburg) and DE-CIX (Frankfurt) attract traffic from landlocked and island economies that lack viable domestic IXPs [Kende, 2020a,b]. The mechanism requires competitive access to the regional hub; where a single operator controls the cross-border path, the hub-and-spoke route is unavailable regardless of the hub’s quality. For small island states and thin markets, hub-and-spoke access to the nearest competitive regional hub is the primary intervention; domestic IXP formation is a second-stage investment once anchor traffic reaches viable thresholds. Operational sequencing guidance is in Section 6.

This extension is qualitatively motivated and consistent with the aggregation logic of the main model; its operational parameters are not yet empirically established. Applying the escape condition from Supplementary Appendix A.5 at regional rather than national scale—aggregating three to five thin-market countries with similar traffic profiles—suggests that a regional hub would need to reach roughly the same 30–50 traffic-weighted member threshold as a national IXP, since the escape condition depends on combined traffic volume presented to the Tier-1 carrier rather than on the number of countries contributing it. This is an indicative range derived from the existing model under regional aggregation assumptions, not an independently calibrated target; TTLs applying the framework in thin-market contexts should treat hub-and-spoke sequencing as a first-order hypothesis supported by qualitative evidence, not as a recommendation with the same evidential grounding as the main aggregation result.

4.4. The Hyperscaler Scope Condition

Hyperscalers (Google, Meta, Amazon) invest in their own submarine cables and edge caches, absorbing a substantial share of consumer traffic [Böttger et al., 2018]. This bypasses the wholesale transit market for consumer content. The framework’s claim is therefore conditional: it applies most forcefully to the *production tier*—traffic for which latency constraints (<100ms round-trip)

or regulatory data-residency requirements make offshore caching technically infeasible or legally impermissible. This tier constitutes 15–25% of total bandwidth but is defined by the economic value of its transactions.¹⁴ Where hyperscaler penetration is high, the primary IXP beneficiaries shift from consumer ISPs to production-tier users—government networks, financial institutions, health systems—implying a different anchor-tenant strategy and evaluation metric. In markets where mobile money and government digital services are nascent, the framework identifies a *target state* rather than a binding current constraint.

5. Empirical Evidence

A Note on Evidence and Identification

The evidence below is mechanism-consistent across multiple independent sources and does not meet the bar for causal identification. The evidence is organized in a single hierarchy: the 2Africa differential treatment effect is the primary evidence most relevant to the paper’s target population; the IXP participant threshold regression and the scope-condition cases are supporting evidence; cross-country regressions are descriptive context explaining why infrastructure alone is an insufficient predictor of prices. The pre-specified MOZIX prospective test (Supplementary Materials, Section S13) is the path toward stronger identification; exogenous variation in IXP establishment would allow causal estimation of magnitude.

5.1. The 2Africa Capacity Shock (Primary Evidence)

In 2023, the 2Africa cable—the world’s largest submarine system, connecting 33 countries—began commercial operations across Sub-Saharan Africa. Its scale makes it the most informative single event in recent internet infrastructure history: a large, exogenous capacity shock landing simultaneously across markets that differed in one observable respect relevant to the framework—whether a carrier-neutral IXP was present at the time of landing. Mozambique is the motivating case.

¹⁴The 15–25% estimate is based on [Sandvine \[2023\]](#) traffic classification data. Sandvine’s classification methodology is proprietary and not independently verified; the range should be treated as an order-of-magnitude indicator rather than a precise measurement. Regulatory data-residency requirements are documented in [Central Bank of Kenya \[2019\]](#), [National Information Technology Development Agency \(NITDA\) \[2019\]](#), and [Republic of Ghana \[2012\]](#). Latency arithmetic is in Supplementary Materials, Section S2.

Two years after the 2Africa landing at Maputo, wholesale prices have not moved—\$4.89/Mbps over six consecutive post-activation quarters— because Telecomunicações de Moçambique controls all three landing stations and MOZIX, the nominally present IXP, was co-located with TDM and carrier-neutral in name only (Figure 8). Supply-contract and integration-delay alternatives both predict a delayed price decline, not a permanent one; comparable landings in competitive gateway markets—SEACOM in Kenya (2009), MainOne in Nigeria (2010)—produced wholesale price responses within the first year, suggesting integration lags do not uniformly prevent response where incumbent control is absent.¹⁵

The 2Africa landing provides a difference-in-differences identification strategy that does not condition on IXP formation: treatment is the cable landing event, and carrier-neutral IXP status is a within-sample moderator. The specification is:

$$\ln P_{it} = \alpha_i + \gamma_t + \beta_1 \cdot \text{Post}_{it} \cdot 2\text{Africa}_i + \beta_2 \cdot \text{Post}_{it} \cdot 2\text{Africa}_i \cdot \text{IXP_CN}_i + \varepsilon_{it} \quad (2)$$

where α_i and γ_t are country and year fixed effects, Post_{it} indicates the period at or after the country-specific 2Africa landing year, and IXP_CN_i indicates carrier-neutral IXP presence at the time of landing. Standard errors are clustered at the country level.

The correct interpretive frame for Table 2 is *mechanism-consistent bounds*, not treatment effects. As Table 3 shows, IXP-CN countries ($n = 6$) were better-governed, wealthier, and had higher ICT development than non-IXP-CN countries ($n = 4$) on every pre-treatment observable. A skeptical reader is right to note that the post-landing price divergence may partly reflect pre-existing institutional advantages rather than the carrier-neutral IXP per se. The robustness specifications in Supplementary Materials, Tables S4–S7—controlling for GDP per capita and ICT development index trajectories—do not resolve this confound; they confirm the direction is consistent but leave the upper-bound characterization intact. What the 2Africa event identifies is therefore not the

¹⁵Both countries’ long-run price trajectories (2016–2025) are in Supplementary Materials, Section S9.1. The prospective MOZIX test—holding cable capacity constant while governance at the exchange point changes—is pre-specified in Supplementary Materials, Section S13. The minimum evaluation window runs Q4 2025 through Q3 2026; the criterion is a sustained excess price decline exceeding 10 percentage points over four or more consecutive post-treatment quarters. At the time of writing the criterion has not been satisfied and no conclusion is drawn.

Table 2: 2Africa Capacity Shock: Price Effects by Carrier-Neutral IXP Status

	(1) All treated	(2) No IXP-CN	(3) IXP-CN present
Post $\times$ 2Africa ($\hat{\beta}_1$)	−0.07 (0.15)	+0.04 (0.18)	—
Post $\times$ 2Africa $\times$ IXP_CN ($\hat{\beta}_2$)	−0.31* (0.17)	—	−0.38** (0.16)
$\hat{\beta}_1 + \hat{\beta}_2$ (IXP-CN total effect)	−0.38** (0.16)		
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
<i>N</i> (country-years)	136	56	80
Countries (treated/control)	10/7	4/7	6/7

Notes: Dependent variable: $\ln(\text{IP transit price, \$/Mbps})$, 2019–2025. Standard errors clustered at country level.
* $p < 0.10$; ** $p < 0.05$. Column (2): no carrier-neutral IXP at landing. Column (3): carrier-neutral IXP present.
Pre-trend test: $p = 0.71$; with $n = 10$ treated countries the test has low power and should not be read as strong reassurance. Full methodology and robustness specifications in Supplementary Materials, Sections S12–S13.

Table 3: Pre-Treatment Observables: IXP-CN vs. Non-IXP-CN Countries at 2Africa Landing

Variable (pre-treatment mean)	IXP-CN ($n = 6$)	No IXP-CN ($n = 4$)	Difference
GDP per capita, PPP (2019, \$000s)	5.2	3.1	+2.1
WGI Government Effectiveness (z-score)	−0.41	−0.89	+0.48
ICT Development Index	4.3	3.1	+1.2
Mobile penetration (% population)	82	61	+21 pp
IP Transit Price, 2019 (\$/Mbps)	3.8	5.1	−1.3

Sources: World Bank WDI, WGI (2019); ITU ICT Development Index (2019); TeleGeography (2019). *Note:* IXP-CN countries were more developed, better-governed, and had lower baseline prices prior to the 2Africa landing on every observable dimension. The −0.38 log-point differential should be treated as an upper bound on the causal IXP-CN effect: some portion of the post-landing price divergence plausibly reflects pre-existing institutional advantages of the IXP-CN group rather than the carrier-neutral IXP per se. Robustness specifications controlling for GDP per capita and ICT development index trajectories are in Supplementary Materials, Tables S4–S7; the direction is consistent across all specifications and the upper-bound characterisation stands.

causal effect of IXP-CN status but the *maximum plausible differential attributable to carrier-neutral aggregation conditional on a large exogenous capacity shock*. That bound is informative: it is the right quantity to assess for a task team deciding whether to invest in aggregation infrastructure alongside a cable project, provided they understand it as a ceiling, not a point estimate.

Table 2 yields three findings. A large, exogenous capacity shock produced no average price response across all 2Africa-connected markets ($\hat{\beta}_1 = -0.07$, Column 1). In countries without carrier-neutral IXPs at landing the point estimate is positive (+0.04, Column 2), directionally consistent with the prediction that capacity without aggregation does not alter the incumbent’s incentive to compete. In countries with carrier-neutral IXPs at landing the price effect is negative and significant ($\hat{\beta}_1 + \hat{\beta}_2 = -0.38$, implied differential ≈ 32 percentage points, Column 3); this estimate is an upper bound, for the reasons stated above, and should be read as indicating direction and rough magnitude rather than precise size. Before-after comparisons from Qatar (−44 pp excess price decline post-IXP formation) and Croatia (−38 pp)—the two cases in the panel with the strongest institutional environments and no documented concurrent policy events—are consistent with the direction of the 2Africa differential in competitive-gateway markets outside Sub-Saharan Africa; full country results are in Supplementary Materials, Table S5.

5.2. IXP Participant Scale Threshold (First Supporting Evidence)

The IXP participant threshold regression uses a supply-side variable—the number of active member networks at the largest IXP in each country—that is not subject to the transit-share circularity: IXP participant count is determined by the investment decisions of member networks, not by the bilateral bargaining outcome the model predicts. Two limitations bound the interpretation before the result is stated. First, the empirical breakpoint will be compared to the calibrated range $N^* \in [25, 35]$ from Supplementary Appendix A.5, but that calibration uses parameters matched to developing-country conditions rather than independently estimated from the price panel; the overlap between the empirical and calibrated ranges is therefore an internal consistency check, not independent corroboration. Second, competitive markets may independently attract more IXP members because they are competitive, not because IXP membership made them competitive; the

regression establishes a cross-sectional association consistent with threshold dynamics, not a causal threshold.

With those bounds in mind: a threshold regression [Hansen, 2000] using IXP participant count as the threshold variable in a regression of $\ln(\text{price})$ on infrastructure controls, with the grid searched over candidate values $\{20, 30, 40, 50, 75, 100, 150, 200\}$ participants, maximises R^2 at a breakpoint of **30 IXP participants** ($R^2 = 0.496$; below-threshold coefficient: 0.838 log points, SE: 0.084, $p < 0.001$). Markets below 30 participants exhibit median prices of \$3.08/Mbps; markets above show \$0.29/Mbps—a $10.5\times$ differential.¹⁶ This breakpoint falls within the calibrated range and is consistent with the model’s aggregation prediction; the operational planning target of 40–50 members reflects adjustment for partial traffic complementarity ($\alpha < 1$) and provides a conservative buffer above the empirical floor.

Countries at the 30-participant boundary have a median transit share of 53.7%—Aggregation Zone (40–70% transit), corresponding to $2 \leq R < 5$ in Table 1—while countries below the threshold have a median transit share of 70.7% (Locked-in Zone, $R \geq 5$). The 30-member breakpoint is thus the IXP scale at which countries in this panel transition from the Locked-in Zone toward the Peering Cliff. These zone labels use transit share as an outcome proxy; structural classification (gateway ownership, IXP governance) is the operationally reliable approach and may differ from transit-share assignments in SOE-dominated markets. Full zone robustness tables and the transit-share typology are in Supplementary Materials, Table S11.

5.3. Scope-Condition Evidence (Second Supporting Evidence)

Proposition 4 predicts that aggregation delivers no price benefit where Conditions (C1)–(C3) fail.

Three independent tests examine this prediction from different angles.

¹⁶Bootstrap 90% CI for the breakpoint: [30, 50] participants (500 iterations, seed: 20260115); 94% of bootstrap samples select 30. Extended grid $\{5, 10, 15, 20, 30, \dots\}$ confirms the lower bound is not a grid-boundary artifact. Full R^2 table in Supplementary Materials, Section S5.

5.3.1. SOE-Dominated Markets: C3 Failure

Two cases provide the primary evidential basis for the SOE scope condition; the cross-sectional regression is directionally consistent but fragile and is presented after the case evidence.

Viet Nam. Three enterprises (VNPT, Viettel, and FPT) collectively control over 90% of international gateway capacity. Despite access to six submarine cable systems—including the 2016–2017 APG system adding 54 Tbps of regional capacity—wholesale prices have remained approximately \$1/Mbps without meaningful decline following capacity additions. FPT is publicly listed but operates under regulatory conditions that constrain its independence as a gateway competitor: Decree 72/2013/ND-CP requires licensed operators to route international traffic through state-designated interconnection points, and FPT’s landing rights are co-licensed with VNPT under joint facility arrangements giving the incumbent effective veto over independent peering.¹⁷ The operative variable for Condition (C3) is regulatory control over gateway access decisions, not formal ownership.

Pakistan. PTCL was 26% privatized in 2006 with a majority stake sold to Etisalat. Wholesale gateway pricing did not decline in the years following; PTCL retained dominance through continued ownership of legacy landing station infrastructure. The competitive mobile sector—Telenor, Jazz, Zong entering 2005–2009—simultaneously achieved rapid retail price declines through competitive entry. The within-country contrast between unchanged wholesale gateway pricing and falling retail mobile prices is consistent with the paper’s argument that competitive entry at the relevant market layer is what disciplines pricing. The comparison is approximate—wholesale international gateway and retail mobile are different markets—but country-level factors are held constant, leaving market-layer competitive structure as the operative difference.

A cross-sectional regression yields an interaction coefficient of 0.815 ($p = 0.047$) when China is included but loses significance when China is excluded ($\beta = 0.590$, $p = 0.16$); China’s Cook’s distance ($D = 0.71$) dominates the full-sample result. The regression is relegated to Supple-

¹⁷FPT competes actively in retail markets; the effectively state-constrained classification applies specifically to gateway access decisions. Full rationale in Supplementary Materials, Section S9.2.

mentary Materials, Table S7. The Viet Nam and Pakistan case evidence is the primary basis for this scope condition; the regression is directionally consistent but should not be treated as independent statistical corroboration.

The near-zero explanatory power of infrastructure in SOE-dominated markets ($R^2 = 0.01$ within the Asian subsample) is consistent with this scope condition. The “Asia” coefficient conflates three structurally distinct groups: competitive East Asian markets (Japan, South Korea, Singapore, Taiwan) consistent with the framework; SOE-dominated markets (China, Viet Nam, Pakistan) where state ownership severs the bargaining channel; and ambiguous Southeast Asian markets (Indonesia, the Philippines) where data coverage is insufficient for a direct test. Removing the three SOE countries reverses the capacity coefficient sign to $\beta = -0.17$ among the remaining 11 Asian markets; restricting to the four competitive East Asian markets yields $\beta \approx 0$, reflecting a price-floor effect in markets already in the Peering Zone. Full subgroup regressions are in Supplementary Materials, Section S7b.

The African and Asian SOE cases differ in mechanism. In China and Viet Nam, state ownership reflects deliberate industrial policy keeping prices strategically low. In African markets—Cameroon, Ethiopia, Angola—SOE control operates through gatekeeping rents under soft budget constraints, producing the opposite outcome: high prices. Table 4 provides operational guidance for distinguishing the two mechanisms.

5.3.2. *Governance as an Alternative Explanation*

Unobserved institutional quality may drive both IXP formation and competitive pricing, making governance rather than aggregation the true determinant of prices. The most direct evidence against the governance-only explanation is cross-country: governance explains essentially none of cross-country price variance ($r = -0.002$), while IXP membership explains a substantial share ($r = -0.785$, $p < 0.05$; Supplementary Materials, Section S6). The Ghana-Nigeria bilateral comparison (Table 5) corroborates this: if governance determined wholesale pricing, Ghana (WGI Government Effectiveness z -score: -0.09 , approximately 55th global percentile) should exhibit

Table 4: Distinguishing Soft Budget Constraints from Security Architecture: Observable Indicators

Indicator	Soft Budget Constraint	Security Architecture
SOE financial position	Operating losses; history of government bailouts; cross-subsidization across segments	SOE may be profitable; revenue used to fund security infrastructure
Regulatory mandate	No explicit routing mandate; prices reflect extraction rents	Legal requirement to route traffic through government-controlled inspection points
Prices relative to costs	Prices high relative to visible infrastructure investment	Prices may reflect genuine inspection costs; high even when capacity is ample
Private sector entry	Entry attempted, prices remain high despite competition in retail market	Entry restricted by law or by mandatory gateway routing requirements
Reform pathway	Competition policy; corporatization; hard budget targets	Requires policy-level decision to separate inspection from commercial transit
Illustrative cases	Cameroon, Angola, Pakistan	China, Iran, possibly Viet Nam (partial)

Notes: Categories are not mutually exclusive. TTLs should consult country specialists and review ITU/Freedom House assessments of internet filtering practices before concluding which mechanism is operative.

lower prices than Nigeria (WGI z -score: -0.85 , approximately 31st global percentile). The opposite holds even after adjusting for market size—Ghana pays approximately $1.86\times$ the Nigerian price after removing estimated scale effects (raw ratio: $2.88\times$). Governance differences are an unlikely sole explanation for this differential, though they cannot be fully excluded as a contributing factor.

A demand-side alternative is equally consistent with the bilateral data: Nigeria's fintech ecosystem (Paystack, Flutterwave, Interswitch) could independently drive both IXP investment and competitive transit pricing. Supply-push (IXPs shifting bargaining leverage) and demand-pull (production-tier demand growth jointly driving IXP formation and price competition) are observationally equivalent in a bilateral cross-sectional comparison. The comparison rules out governance as the *sole* determinant; it cannot identify the dominant channel between supply-push and demand-pull.

Table 5: The Ghana-Nigeria Governance Test

	Ghana	Nigeria
<i>Governance Indicators (WGI)</i>		
Government Effectiveness (z-score)	−0.09	−0.85
Global percentile rank	~55th	~31st
<i>Market Size</i>		
Population (millions)	~34	~220
Internet users (millions, 2023)	~16	~103
<i>Aggregation Infrastructure</i>		
IXP members	36	120+
Peak traffic (Gbps)	~50	~325
<i>Outcome</i>		
IP Transit Price (2024)	\$1.18/Mbps	\$0.41/Mbps
Price ratio (raw)		2.88×
Price ratio (population-adjusted)[‡]		1.86×

Sources: World Bank WGI (2023), ITU, PeeringDB, TeleGeography (2024).

[‡] Adjusts for scale economies using $\hat{\gamma} = 0.21$ (OLS coefficient on $\ln(\text{population})$, Table 7, Col. 2).

5.3.3. *IXP Formation in the Target Population: Scope Confirmation*

A structural feature of the before-after sample limits its applicability to the paper’s primary target population. The 14-country sample is selected on IXP formation: only countries where an IXP was successfully established appear in the data. Countries where incumbent carriers blocked IXP entry—the locked-in markets the analysis is designed to diagnose—are absent by construction. This selection introduces upward bias: the sample systematically excludes the worst-performing cases. For a TTL evaluating IXP investment in a currently locked-in market, the 14-country before-after results are not the relevant evidence base; they describe what happened in markets that had already overcome the blocking problem.

The 2Africa differential (Section 5.1) does not share this selection problem. The 2Africa sample includes all ten connected countries regardless of whether they had a carrier-neutral IXP at landing—treated and control units are assigned by geography, not by prior IXP success. This is why the 2Africa estimate is the primary evidence for the Sub-Saharan African target population, while the before-after table serves as scope-condition illustration rather than mechanism evidence.

Within the 14-country sample, the Sub-Saharan Africa and Central Asia subset is consistent with the scope condition rather than the mechanism. Once the Cameroon data artifact is excluded (pre-IXP price of \$73.36/Mbps exceeds Angola—among the world’s most expensive markets—by a factor of 4.3, and is almost certainly a thin-market reporting artifact), the two-country median excess decline is +1.1 percentage points (Kazakhstan: –11.2 pp; Uzbekistan: +13.4 pp), indistinguishable from zero.¹⁸ Kazakhstan’s IXP operates in a partially competitive gateway market (–11.2 pp); Uzbekistan’s does not, and the positive excess reflects the Proposition 4 prediction that IXPs without competitive gateway access deliver no price benefit.

5.4. Cross-Country Patterns (Descriptive Context)

The cross-country dataset covers 169 economies, with IP transit prices ranging from \$0.09/Mbps (Netherlands) to \$60/Mbps (Pacific island states), median \$0.60/Mbps (Table 6). Countries with

¹⁸Full 14-country results are in Supplementary Materials, Table S5.

IXPs exhibit median prices of \$0.52/Mbps versus \$1.41/Mbps without—a $2.7\times$ difference—but this raw gap is not evidence of an IXP effect: countries with IXPs are systematically higher-income and better-connected, and the comparison does not control for development level or infrastructure endowment.

Table 6: Summary Statistics: Cross-Country Sample

Variable	N	Mean	Median	Min	Max
IP Transit Price (\$/Mbps)	97	4.21	0.60	0.09	60.00
Int'l Capacity (Gbps)	169	64,205	4,842	4	1,274,016
IXPs	169	4.1	1	0	134

Source: TeleGeography, PeeringDB (2024–2025). Coverage is skewed toward medium and large markets; SIDS and fragile states are underrepresented. IP transit price distribution is highly right-skewed (mean $7\times$ median); all regressions use log specifications.

Cross-country infrastructure regressions have weak explanatory power on their own—but that weakness is the finding. Adding regional fixed effects to a developing-country regression raises R^2 from 0.035 to 0.350 (Table 8): the dominant source of price variation is not how much infrastructure a country has but where it sits institutionally. This result motivates the emphasis on market structure over capacity: infrastructure effects on prices operate *through* governance and bargaining arrangements, which vary primarily along regional and institutional rather than physical dimensions.

Asia exhibits $7\times$ higher prices at equivalent infrastructure relative to Latin America ($p < 0.001$), driven by the SOE-dominated subgroup discussed in Section 5.3; Africa's coefficient is positive but not significant, consistent with heterogeneous gateway governance across African markets. The cross-country sample covers 97 of 169 economies; missing countries are disproportionately small island states and fragile states—precisely the markets the analysis most aims to inform—and the estimated associations should be read as upper bounds on expected effects in the most fragile markets.

Endogeneity note. The traffic asymmetry ratio (R) is likely endogenous: high transit prices suppress local content hosting, which increases asymmetry, which sustains high prices.

Table 7: Determinants of IP Transit Prices: Key Results

	Dependent Variable: $\ln(\text{IP Transit Price, \$/Mbps})$			
	(1) <i>Bivariate baseline</i>	(2) + Capacity	(3) <i>Preferred</i> + Regional FE	(4) Comple- mentarity test
IXP Presence (0/1)	−1.80*** (0.31)	−0.16 (0.28)	−0.18 (0.25)	−2.14* (1.20)
$\ln(\text{International Capacity})$		−0.63*** (0.08)	−0.41*** (0.11)	−0.49*** (0.15)
$\text{IXP} \times \ln(\text{Capacity})$				0.30* (0.16)
Regional Fixed Effects	No	No	Yes	Yes
Observations	97	97	97	97
R^2	0.11	0.38	0.75	0.75

Robust standard errors (HC1) in parentheses. *** $p < 0.01$, * $p < 0.10$. *Note:* Column (3) is the preferred specification. The binary IXP indicator is a coarse proxy; the model predicts that *scale and carrier neutrality* matter, not presence alone—the threshold regression in Section 5.2 is the operationally stronger evidence. Column (4) tests complementarity: the positive, marginally significant interaction ($p = 0.059$) indicates the IXP effect is attenuated in high-capacity markets, consistent with cable serving as a background enabling condition. Full specifications in Supplementary Materials, Section S7.

Table 8: Regional Heterogeneity in the Infrastructure-Price Relationship

	Dependent Variable: ln(IP Transit Price)	
	(1) Without Regional FE	(2) With Regional FE
ln(International Capacity)	−0.08 (0.11)	−0.21 (0.14)
IXP Present	−0.31 (0.28)	−0.18 (0.24)
<i>Regional Fixed Effects (vs. Latin America):</i>		
Asia	—	1.99*** (0.42)
Middle East	—	0.95** (0.46)
Africa	—	0.48 (0.34)
Observations	52	52
R^2	0.035	0.350

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$. Notes: Sample restricted to developing countries with price data. Latin America is the reference category. Full within-region specifications are in Supplementary Materials, Section S7.

This feedback loop implies that the associations reported above are likely *attenuated*. It also characterizes a bad equilibrium from which incremental interventions may be insufficient to escape—the theoretical basis for coordinated multi-instrument approaches in markets with high transit-share indicators (used here as an outcome proxy for R ; see Table 1). The loop is a conjecture consistent with the static Nash model but does not follow formally from it; a dynamic extension establishing multiple equilibria is left for future work.

Capacity control: panel evidence. The cross-country results above use a binary IXP indicator and cross-sectional capacity measures. Supplementary Materials, Section S7b, reports a country-year panel specification using TeleGeography Transport Networks data on total used international capacity (K_{ct} , Gbps, 2010–2024). When $\ln(K_{ct})$ is included as a continuous capacity control with region and year fixed effects, the continuous IXP participant measure is absorbed (96% attenuation, $p = 0.93$)—consistent with capacity and IXP participant count trending upward together. However, the IXP threshold indicator ($\mathbb{1}[\text{IXP} \geq 30]$) survives at -0.99 log points ($p < 0.001$), approximately 63% lower prices, even with capacity fully controlled. This is more consistent with the Nash bargaining model than a smooth linear gradient would be. The model predicts a *discontinuity* where bargaining leverage shifts, not a smooth relationship between member count and price. The capacity controls show that the empirical IXP effect is concentrated at the structural break the model derives, not distributed across the participant count distribution. The capacity $\times$ IXP interaction is positive ($+0.11$, $p = 0.004$), indicating that capacity and IXP governance are partial substitutes for domestic traffic routing. Where the IXP has already solved the domestic routing problem, additional international bandwidth has diminishing returns. Replication: `10_capacity_control_regressions.py`.

Locked-in Markets. Despite global price convergence, 21 countries remain above 95% transit share (classified by transit share, an outcome proxy; structural verification recommended before applying intervention pathways). These are not homogeneous: approximately 8 are SOE-gateway dominated, approximately 6 are thin-market economies where demand constraints are co-binding, and approximately 7 have competitive gateway structures but underdeveloped aggregation

infrastructure. TTLs should verify the operative mechanism before selecting an intervention pathway.

What the evidence supports. Across three independent sources—the 2Africa capacity shock, the IXP participant threshold regression, and the scope-condition cases—the pattern is consistent: cable capacity alone does not compress prices; carrier-neutral aggregation at sufficient scale does, but only where competitive gateway access and market-responsive pricing are already present. None of the evidence meets the bar for causal identification, and each source has limitations documented above. The policy implication does not require causal precision to be actionable, however. A task team does not need to know the exact magnitude of the IXP effect to know that financing cable capacity in Mozambique without addressing landing-station governance was a diagnostic error. The framework’s value is in identifying *which constraint is binding*—a question the evidence answers consistently across sources—not in estimating the welfare magnitude of resolving it. The policy and operational guidance in Section 6 is calibrated to this evidential standard.

6. Policy and Operational Implications

6.1. Why Structural Remedies Outperform Conduct Remedies

Cost-of-service access regulation requires the regulatory capacity to compute cost-based prices and audit compliance. That capacity is typically absent in the markets where incumbents have successfully blocked IXP formation [[Estache and Goicoechea, 2009](#), [Laffont and Tirole, 2000](#)]. Aggregation works differently: it disciplines pricing through market structure rather than regulatory command. Transparency mandates (requiring Reference Interconnection Offer publication) are low-cost complements to aggregation: they generate information without sustained enforcement capacity. Hard price regulation should be sequenced as a last resort where structural remedies have demonstrably failed.

6.2. Policy Ranking

Aggregation outperforms mandated bill-and-keep across a wide range of parameter values: bill-and-keep efficiency is contingent on traffic symmetry, and mandating it between asymmetric networks

creates incentives for de-peering or quality degradation [Wright, 2002, Gans and King, 2001]. The general theory of access pricing establishes that efficient interconnection arrangements must account for cost asymmetries and outside options [Armstrong, 2002]; in the Global South context, these asymmetries are severe enough that regulatory mandates operating independently of market structure are unlikely to produce competitive outcomes. IXP investment creates the symmetry that makes settlement-free exchange viable; aggregation must therefore often precede rather than follow mandated peering. Pass-through from lower wholesale prices to retail consumers depends on retail market structure and is not estimated here.

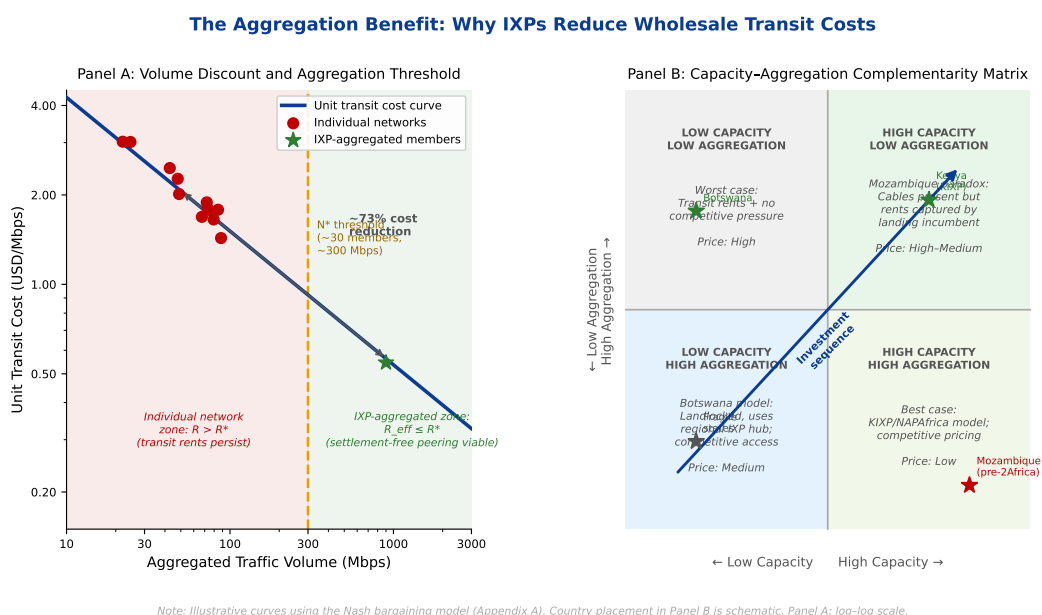


Figure 3: Volume Discount Curve—The Aggregation Benefit. Unit transit costs decline steeply with aggregated volume.

6.3. Retail Pass-Through: A Necessary Condition for Welfare Impact

The framework operates at the wholesale level: carrier-neutral IXPs at sufficient scale compress wholesale IP transit prices. Whether these savings reach end users depends on retail market structure. Where the gateway incumbent also dominates retail connectivity, the margin improvement need not pass through to consumers; the welfare case for IXP investment is therefore strongest in competitively structured retail markets [Genakos and Valletti, 2011, Briglauer et al., 2019]. Evaluation

metrics for IXP investments should include retail price tracking alongside wholesale benchmarks—a wholesale decline without a corresponding retail movement is diagnostic of downstream market structure and should trigger a second-stage assessment.

The concern is sharpest in Sub-Saharan Africa. There, the same incumbent that controls the landing station frequently also dominates fixed broadband retail. The vertical integration that makes gateway governance the bottleneck at the wholesale layer may simultaneously suppress retail pass-through at the downstream layer. Where this vertical structure is present, IXP investment addresses the wholesale constraint but leaves the retail constraint intact. A complete welfare assessment requires separate evaluation of retail competitive conditions. In markets where mobile operators have entered retail broadband competitively, as in Kenya, Nigeria, and Rwanda, competitive pressure at the retail layer increases the likelihood that wholesale savings reach end-users. This strengthens the welfare case for aggregation investment. I do not estimate retail pass-through rates. The pass-through condition is a scope restriction on the welfare claims, not a critique of infrastructure investment (full guidance in Supplementary Materials, Table S0b).¹⁹²⁰

6.4. Regime Classification and Intervention Pathways

Table 9 classifies markets by framework applicability and maps each to an intervention pathway. TTLs should use IXP participant count as the primary operational diagnostic and validate zone classifications against structural indicators (gateway ownership, IXP governance, landing-station access conditions). In Africa, core recommendations apply directly where gateway markets are

¹⁹A distinct but complementary constraint is the “usage gap”: even where wholesale savings pass through to retail prices, end-users may lack internet-enabled devices. Demand-side guarantee facilities—such as small-loan guarantees backing smartphone financing plans offered by mobile operators or microfinance institutions—address this constraint but operate at a different layer than the wholesale interconnection channel I analyze. The World Bank deploys targeted instruments of this kind in several IDA-eligible markets; they are complementary to, not substitutes for, aggregation investment.

²⁰The “waterbed effect” documented in mobile termination rate regulation—where mandated wholesale price reductions are partially offset by retail price increases [Genakos et al., 2018]—is a potential concern for any wholesale pricing intervention. Here the channel differs: IXP aggregation reduces wholesale transit prices by changing market structure rather than by regulatory mandate, so the incumbent’s pricing power is diminished rather than redirected. However, where the incumbent is vertically integrated—controlling both the landing station and retail broadband—the vertical analog of the waterbed is operative: wholesale savings may be absorbed into the incumbent’s retail margin rather than passed through to end-users. This is the concern addressed by Pre-condition 1 above. The waterbed risk is greatest where the analysis already recommends separate retail structure assessment before projecting wholesale gains to consumer welfare.

structurally competitive. In SOE-dominated markets (§5.3), structural reform must precede infrastructure investment; effective reform need not require full privatization—gateway demonopolization, corporatization with hard budget targets [Klien, 2014, Shleifer and Vishny, 1994], and regulatory separation each introduce competitive discipline without ownership transfer. This sequencing recommendation rests on the Pakistan and Viet Nam case evidence (Section 5.3) and the Kornai soft-budget-constraint literature; the cross-sectional statistical interaction is directionally consistent but fragile (Supplementary Materials, Table S7), and TTLs should confirm the operative mechanism before applying this pathway. In Latin America, focus should shift to underserved areas and regional integration.

6.5. Operational Instruments

For markets in the Locked-in Zone ($R \geq 5$, an operational judgment about $N^*(R)$ scale requirements rather than a model-derived boundary), standalone instruments are unlikely to succeed. The coordinated package combines regulatory foundation (DPF), physical infrastructure (IPF), and private investment mobilization (IFC/MIGA), addressing the coordination failure that keeps markets locked in suboptimal equilibria.

Conditioning de-risking on interconnection governance. The World Bank Group’s consolidated Guarantee Platform [World Bank Group, 2024b] provides political risk insurance, credit enhancement, and concessional blended finance that lower the weighted average cost of capital for private operators in frontier ICT markets. These instruments address a real barrier—the combination of political risk, currency mismatch, and thin markets that makes standalone private investment in digital infrastructure unviable in IDA-eligible countries. But there is a structural limitation: De-risking instruments that protect a fiber or data center investment without conditioning on interconnection governance replicate the same necessary-but-not-sufficient problem that applies to cable capacity. A MIGA guarantee that lowers the cost of capital for a fiber operator in a market where the incumbent controls the landing station reduces investment risk without addressing the wholesale pricing bottleneck. The operational implication is that IFC equity positions, MIGA guarantees, and IDA PSW facilities in digital infrastructure should be

Table 9: Regime Classification: Framework Applicability by Market Characteristics. Market type classifications are based on structural indicators (gateway ownership, IXP governance), not transit-share zones; for the zone panel, see Supplementary Materials, Table S11.

Market Type	Framework Applies?	Evidence Strength	Pass-through Risk [‡]	Intervention Pathway
Competitive, weak IXP (e.g., Mozambique, pre-reform Kenya)	Yes	Strong	Medium	Carrier-neutral aggregation investment
Competitive, with IXP (e.g., Nigeria, Ghana)	Partial	Moderate	Low	Extend benefits; regional integration
SOE-dominated, soft constraints (e.g., Viet Nam, Cameroon) [†]	Limited	Moderate	High	Competition at gateway must precede infrastructure
SOE-dominated, hard constraints	Partial	Weak	High	Aggregation may work if managerial incentives align
Small island states	Limited	Weak	Medium	Geographic constraints dominate; regional hub-and-spoke pooling is the primary intervention (see hub-and-spoke corollary)
Already competitive (e.g., Brazil, South Africa)	No	Strong	Low	Focus on underserved areas

Notes: “Strong” = regression + case evidence. “Moderate” = case evidence or sensitive to specification. “Weak” = limited direct evidence. [†]Indonesia and the Philippines exhibit SOE presence at the gateway but retain elements of competitive discipline; they are classified separately from soft-constraint SOE markets. [‡]Pass-through risk indicates the likelihood that wholesale price reductions will *not* reach end-users due to retail market concentration. High = gateway incumbent also dominates retail; welfare claims require separate retail structure assessment (Pre-condition 1). Medium = retail competition present but incomplete; monitor retail prices alongside wholesale benchmarks. Low = competitive retail market; welfare case applies directly.

Table 10: Operational Instrument Mapping

Intervention	Instrument(s)	Illustrative Operation
Carrier-Neutral Data Centers	IFC ²¹ Equity; MIGA	Teraco (South Africa); MainOne MDXi (Nigeria)
Regional IXP Support	IPF ²² ; Technical Assistance	RCIP ²³ (East Africa); WARCIP (West Africa)
Regulatory Reform	DPF prior action; DDP ²⁴	RCIP DPO (RIO publication); open-access landing station regulation; spectrum allocation advisory
Private Sector De-risking ²⁵	IDA PSW (MGF, LCF); MIGA PRI	IXP/data center investment in IDA-eligible markets; BRIDGE Nigeria SPV structure
Coordinated Package	DPF + IPF + IFC/MIGA + PSW	DPF for policy reform, IPF for infrastructure, IFC/MIGA for private investment, PSW for frontier de-risking. Example: RCIP series; BRIDGE

conditioned on verifiable interconnection governance milestones—carrier-neutral co-location, non-discriminatory peering terms, Reference Interconnection Offer publication—to ensure that de-risked private capital addresses the bottleneck rather than the background enabling condition. BRIDGE Nigeria’s open-access wholesale SPV structure, in which the majority privately-owned vehicle provides non-discriminatory capacity to all licensed operators [World Bank, 2025a], illustrates this pairing: the de-risking instruments and the governance design work jointly with the wholesale pricing channel the model identifies.

6.6. TTL Diagnostic Framework

Practitioners seeking a self-contained operational reference should consult Supplementary Materials, Section S10, which consolidates the regime classification, pre-conditions checklist, failure-modes diagnostic, and instrument-mapping into a single two-page exhibit designed for project preparation use. The discussion below summarises the conceptual basis for that guidance.

A task team should classify a country into one of three regimes using *structural indicators only*—traffic ratio, landing-station market structure, and IXP governance and scale. Wholesale price levels then serve as a *validation check* for the classification, not as an input criterion. If a country structurally resembles a locked-in regime but exhibits low prices, this signals the presence

of additional policy instruments (fiscal subsidies, hyperscaler cross-subsidies, or deliberate SOE pricing policy) that are outside the scope of this diagnostic. The classification and the intervention recommendation should never depend on the variable—price—that the intervention is intended to improve.

Regime Classification. Table 9 maps structural inputs (traffic asymmetry ratio, landing-station ownership, IXP governance and scale) to three regimes with distinct intervention pathways; typical price ranges serve as consistency checks, not classification criteria.

Decision Rules by Regime.

- *Peering Regime:* Do not prioritize new IXP or cable financing absent a clear underserved-region rationale. Focus on demand-side constraints—device penetration, digital literacy, application development. Price regulation is largely redundant.
- *Aggregation Regime:* Prioritize carrier-neutral IXPs and data centers with anchor-tenant coordination. Consider mild conduct remedies (transparency mandates, Reference Interconnection Offers) only once IXP participation is broad and dominant carriers are active members. Avoid hard price regulation before traffic asymmetry has been reduced.

Pre-condition 1 (Retail Market Structure) must be verified before projecting wholesale price reductions through to end-user welfare. The key heuristic: where the gateway incumbent also dominates fixed broadband retail—as in many Sub-Saharan African markets where vertical integration runs from landing station through retail—resolving the wholesale bottleneck via an IXP may transfer rents from Tier-1 global carriers to the domestic retail monopolist, with zero welfare gain for end-users. Conversely, where mobile operators have entered retail broadband competitively—Kenya, Nigeria, Rwanda are the clearest cases in the primary application domain—competitive pressure at the retail layer substantially increases the probability that wholesale savings pass through. These two patterns are observable from market structure data before project appraisal and should be treated as a first-pass welfare filter: in monopoly-retail markets, treat wholesale price targets as intermediate indicators and assess downstream structural competition separately; in competitive-retail markets, the welfare

case applies directly. A wholesale price decline without a corresponding retail movement is diagnostic of vertical foreclosure, not a failure of the aggregation mechanism.

- *Locked-in Regime*: Do not lead with cable or IXP financing alone. Sequence structural reforms—open landing stations, break transit monopolies, or establish regional hubs—before or concurrent with aggregation investment.

The intervention pathway diverges sharply depending on which SOE mechanism is operative, and this divergence must be made explicit before project design begins. In *soft-budget-constraint* markets (Cameroon, Angola, Pakistan: Table 4), the gateway monopoly reflects rent-seeking under weak competitive discipline; competition at the gateway, corporatization with hard budget targets, or regulatory separation can harden incentives and make aggregation effective. Standard development finance instruments—DPF prior actions conditioning on gateway opening, IPF for IXP infrastructure—are applicable. In *security-architecture* markets (China, Iran, and potentially Viet Nam on this dimension: Table 4), high prices and gateway bottlenecks are deliberate instruments of state control rather than economic inefficiencies. In these markets, the standard lending-instrument toolkit is mismatched to the actual constraint: lowering the price is not the sovereign’s objective, and aggregation investment that threatens surveillance architecture will face legal or administrative resistance independent of its economic merits. Task teams should consult the Table 4 indicators and country specialists before committing to a reform-first strategy; where security-architecture features are present, the operational recommendations developed here do not apply.²⁶

Three inputs are sufficient for a first-pass classification, all obtainable at low cost: TeleGeography price benchmarks for context; simple traffic ratio estimates from operators or transit share data; and a qualitative assessment of who controls landing stations and under what access conditions. Before applying the regime classification, five pre-conditions should be verified: (1) retail market

²⁶The recommendation that competition at the gateway precede infrastructure investment in SOE-dominated markets is grounded in the Pakistan and Viet Nam case evidence (Section 5.3) and the Kornai soft-budget-constraint literature. The cross-sectional regression corroborates direction but loses significance when China is excluded from the 28-country sample ($p = 0.16$). This recommendation should be treated as grounded inference from case evidence and theoretical reasoning, not as a statistically established regularity.

structure is sufficiently competitive for wholesale savings to pass through; (2) supply rather than demand is the binding connectivity constraint; (3) a minimum local digital enterprise base exists to anchor IXP participation; (4) the operative mechanism is identified (soft budget constraint vs. security architecture—see Table 4); and (5) the political economy is tractable. Full pre-condition definitions and a structured verification checklist are in Supplementary Materials, Section S10.

The full diagnostic, including a decision matrix and expanded guidance, is in Supplementary Materials, Section S10.

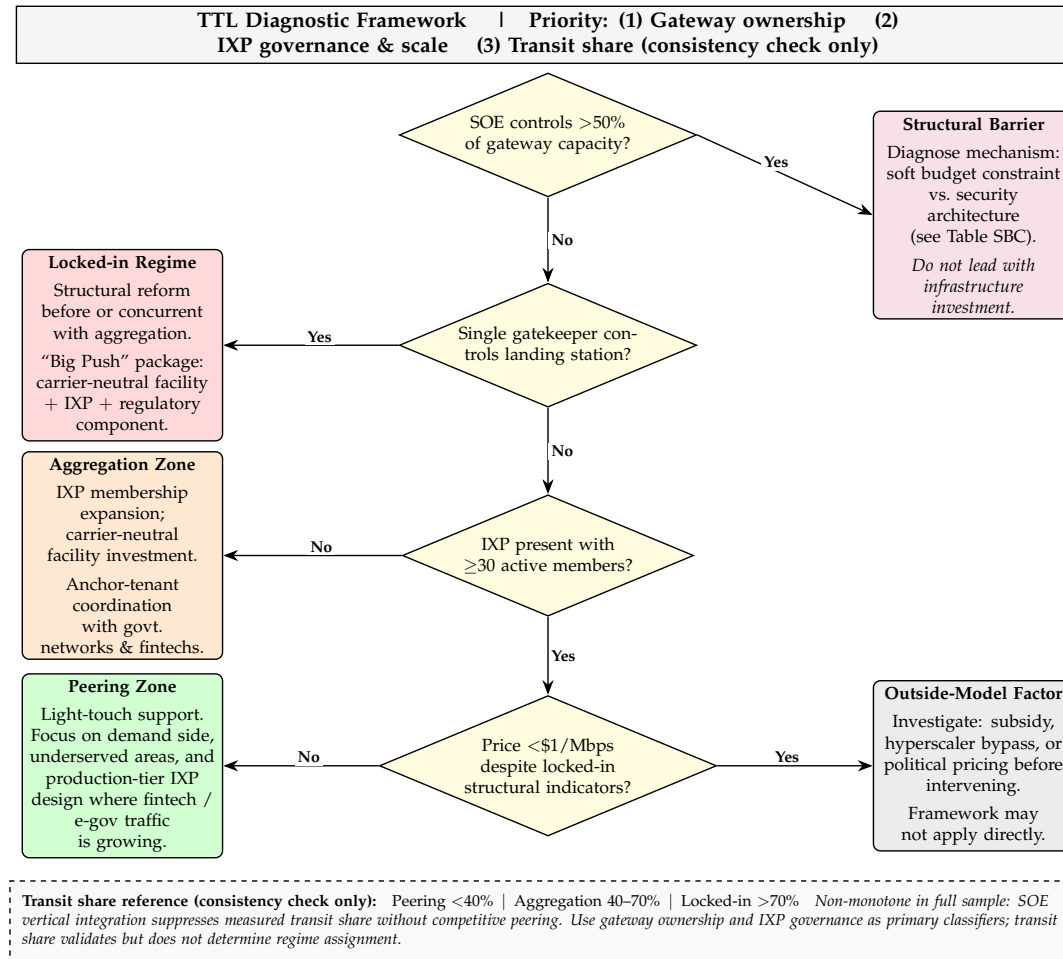


Figure 4: **TTL Diagnostic Framework.** Classify by structural indicators in priority order: gateway ownership first, then IXP governance and scale, then transit share as a consistency check. The transit-share zone typology is non-monotone in the full sample (Supplementary Materials, Table S11); the IXP participant count gradient (Supplementary Materials, Table S11) is the more reliable primary diagnostic. For the full diagnostic table when R is unavailable, see Supplementary Materials, Section S10.

6.7. When Framework Predictions Fail

IXP investment does not guarantee price reductions. Three failure modes account for most documented cases:

1. *Incumbent non-participation or de-peering.* The IXP cannot aggregate traffic if the dominant carrier refuses to participate. **Diagnostic:** Is the largest carrier an active IXP member? **Response:** Regulatory mandates for carriers above a size threshold, or carrier-neutral facilities adjacent to incumbent landing stations.
2. *Insufficient scale.* IXPs require minimum viable membership and traffic (~ 1 Gbps peak) to shift bargaining leverage. The empirical structural break falls at 30 participants; the operational planning target of 40–50 members reflects adjustment for partial traffic complementarity ($\alpha < 1$) and provides a more conservative buffer. However, this target carries substantial uncertainty: across the plausible range of the complementarity parameter $\alpha \in (0.5, 1.0)$ and typical developing-country traffic profiles, the minimum viable membership threshold spans approximately 20 to 60 members. Practitioners should treat 40–50 as a central planning estimate, not a precise threshold. *Critically, the 40–50 member target should be assessed using traffic-weighted membership, not raw member counts.* The escape condition depends on the combined traffic volume the IXP can present to a Tier-1 carrier, not on the number of members per se: an IXP with 35 members that includes anchor tenants generating $20\times$ average traffic volumes may already satisfy $N^* < 35$ in traffic-weighted terms, while an IXP with 50 small-network members contributing near-equal traffic may not (Supplementary Supplementary Appendix A.5, heterogeneous-network remark). Practitioners assessing proximity to the escape condition should calculate the traffic-weighted effective membership $N_{\text{eff}} = \sum_k (Q_{ji}^k / \bar{Q}_{ji})$ and compare it to the 40–50 target, rather than counting members directly. Operationally, IXPs reaching 20 members should trigger active monitoring of transit price responses; IXPs reaching 30 members should trigger anchor-tenant outreach to the largest unconnected domestic networks. The threshold will differ across markets with different traffic profiles, port costs, and carrier participation rates. **Diagnostic:** Does the IXP meet

scale thresholds? **Response:** Anchor tenant coordination (government networks, financial institutions).

3. *IXP governance capture*. An IXP controlled by or co-located with the incumbent cannot provide the independent aggregation required for bargaining leverage to shift. The Mozambique case illustrates this pattern. **Diagnostic:** Is the IXP genuinely carrier-neutral? **Response:** Investment in carrier-neutral facilities distinct from incumbent infrastructure.

Projects should include: (i) provisions ensuring carrier neutrality; (ii) anchor tenant commitments; (iii) regulatory backstops addressing incumbent non-participation. See Supplementary Materials (Section S8) for a detailed troubleshooting checklist.

A note on political economy tractability. The three failure modes above each require regulatory or investment responses that presuppose a tractable political economy—that reform advocacy will not be blocked by incumbents with sufficient leverage to neutralize the instrument (Pre-condition 5). Where the incumbent’s gateway position is its primary remaining competitive asset, project design should anticipate resistance and build in regulatory backstops with credible enforcement, or sequence the intervention to reduce the incumbent’s dependence on gateway rents before pressing for open access.

6.8. Evidence Summary

Table 11 maps each policy recommendation to its primary evidence and scope conditions. Strength grades follow the convention in the supplementary materials: *Moderate* denotes theory plus case evidence or a specification-sensitive regression; *Weak* denotes theoretical prediction without supporting statistical evidence. The full evidence-to-recommendation mapping with evidence-basis codes (statistical / case / theoretical) is in Supplementary Materials, Table S0b.

Cloud Infrastructure Convergence. Submarine cable landing stations, IXPs, and cloud on-ramps are increasingly co-located in carrier-neutral facilities. Current hyperscaler investment patterns systematically favor markets with competitive interconnection infrastructure [Google Cloud, 2024, Amazon Web Services, 2024]. Markets stuck at the Peering Cliff are therefore also those least likely to attract cloud on-ramp investment, suggesting the digital infrastructure gap and the

Table 11: Evidence-to-Recommendation Mapping (Condensed). Full version with evidence-basis codes in Supplementary Materials, Table S0b.

Recommendation	Key Evidence	Strength	Scope Condition
Combine cables with carrier-neutral IXP; target 40–50 member scale (empirical structural break: 30 members; 40–50 is the planning target adjusted for $\alpha < 1$)	2Africa: IXP-CN countries –32 pp vs. no response in non-IXP-CN (Table 4); monotone IXP participant gradient (Table 7); Kenya/Nigeria long-run trajectories (–28–30 pp excess, 2016–2025) ^a	Moderate	Prop. 4 (C1)–(C3) all hold; competitive gateway access confirmed
Prioritize locked-in countries first	21 countries at >95% transit share (outcome proxy; structural verification required); before-after SSA/CA median +1.1 pp (indistinguishable from zero); 2Africa differential is primary mechanism evidence for SSA	Moderate	Country in Locked-in or Aggregation zone; gateway not SOE-controlled
Sequence structural reform before infrastructure in SOE markets	Viet Nam APG non-response; Pakistan within-country comparison; SOE interaction $\beta = 0.59$, $p = 0.16$ (China excluded; Supplementary Materials, Table S7)	Moderate (direction); Weak (precision)	C3 fails; confirm operative mechanism (rent-seeking vs. security architecture) via Table 4 before committing
Production tier requires carrier-neutral local exchange	KIXP latency: 200–600 ms $\rightarrow$ <30 ms; CBK, NDPR, Ghana DPA data-residency mandates; direct welfare measurement unavailable	Moderate	C1 + C2 hold; not applicable in markets too thin to sustain viable exchange
Infrastructure diversification for locked-in markets	Theoretical mechanism only; no systematic portfolio evidence	Weak	C1 fails or C2 fails; treat as last resort pending further evidence

Strength grades: “Moderate” = theory plus case evidence, or specification-sensitive regression; “Weak” = theoretical prediction without supporting statistical evidence. ^aKenya (KIXP, 2000) and Nigeria (IXPN, 2009) predate the TeleGeography panel; trajectories from Supplementary Materials, Section S9.1.

cloud access gap may be converging. See Supplementary Materials (Section S2) for guidance by market tier.

6.9. Alignment with Current World Bank Digital Infrastructure Programming

The World Bank’s recent digital infrastructure commitments target the physical access and demand layers: fiber backbone, electricity, school connectivity, digital literacy. Wholesale interconnection governance receives less systematic attention. The commitments address real constraints. But a structural layer between submarine cable capacity and retail affordability, the layer I analyze, falls outside current programming.

Coverage targets without pricing mechanisms. The Cotonou Declaration on Digital Transformation in Western and Central Africa commits signatory states to broadband coverage for 90 percent of the population and affordable internet access for 70 percent of rural households by 2030 [[Ministers of Digital Economy of the Member States of West and Central Africa, 2025](#)]. These are coverage targets, not pricing targets. The distinction matters. Proposition 4 predicts that coverage expansion without carrier-neutral aggregation will produce heterogeneous pricing outcomes: countries past the Peering Cliff threshold will translate coverage into affordable access, while those that have not will either require sustained subsidy or fail to meet targets because operators cannot profitably serve rural areas at prevailing wholesale costs. The Declaration does include a commitment to “deploy regional Internet exchange points (IXPs) to secure and optimize African data traffic,” but this language frames IXPs as traffic-management tools rather than as aggregation infrastructure that shifts wholesale bargaining leverage.

Backbone investment as a natural experiment. The BRIDGE Project in Nigeria—\$500 million in IDA financing to extend the national fiber backbone from 35,000 to 125,000 kilometers through a majority privately-owned Special Purpose Vehicle [[World Bank, 2025a](#)—offers the closest operational parallel. Nigeria has an established IXP (IXPN, founded 2009) and competitive retail broadband markets, satisfying Conditions (C1)–(C3). The model makes a testable prediction. the \$500 million backbone investment will compress retail prices *if and only if* IXPN operates as a genuinely carrier-neutral exchange and landing station access remains competitive. If

dominant operators can control peering terms or degrade interconnection quality at the exchange, the backbone becomes necessary but not sufficient, the pattern documented in the 2Africa evidence for markets without carrier-neutral IXPs. BRIDGE’s open-access wholesale design aligns with the structural-remedy recommendation developed above; the operation is a live test of whether backbone expansion paired with open-access governance delivers retail price reductions.

Demand-side programs and wholesale cost heterogeneity. The World Bank–IDB “Connected Schools for All” initiative commits \$512 million to connect 3.5 million students across 16 Latin American and Caribbean countries [[World Bank and Inter-American Development Bank, 2023](#)]. School connectivity is a procurement program: schools need bandwidth, and bandwidth has a wholesale price. The wholesale pricing analysis explains why identical investments buy different quantities of bandwidth depending on whether a country has crossed the Peering Cliff threshold. In markets with established carrier-neutral IXPs—Brazil’s IX.br is among the world’s most successful—procurement efficiency is high. In markets without equivalent aggregation infrastructure, the same dollar buys less bandwidth at higher unit cost. None of this argues against school connectivity investment. But procurement cost-effectiveness varies with wholesale pricing conditions, and wholesale pricing conditions vary with interconnection structure.

Gender inclusion and the affordability assumption. The World Bank Group Gender Strategy 2024–2030 targets 300 million additional women using broadband by 2030 [[World Bank Group, 2024a](#)]. The complementary Mission 300 initiative targets electricity access for 300 million people in Sub-Saharan Africa by 2030 [[World Bank Group and African Development Bank, 2024](#)]—a genuine precondition for broadband, and one this analysis does not contest. But the gender broadband target implicitly assumes that once physical access, electricity, and digital literacy exist, affordability follows. It need not. The Nash bargaining model identifies the markets where affordability does not follow from access: those with high traffic asymmetry and no carrier-neutral IXP. A woman in Maputo with a smartphone and electricity still faces data prices high enough to suppress usage if MOZIX does not achieve carrier-neutral governance—the condition whose fulfillment is the subject of the prospective test described in Section 5.1.

Fragile states and the security-architecture distinction. The World Bank’s Accelerating Digital Transformation in FCS program has supported 23 digital operations with approximately \$2.7 billion in financing since 2018 [[World Bank, 2025b](#)]. FCS markets are overrepresented among what the analysis classifies as security-architecture market structures (Table 4), where the state controls telecommunications infrastructure for surveillance or political reasons rather than because of soft-budget-constraint inefficiency. Scope condition (C3), market-responsive carrier pricing, fails by design in these markets. Standard IDA lending instruments (technical assistance, regulatory reform, PPP frameworks) presuppose that the constraint is capacity or institutional knowledge; in security-architecture markets, the constraint is political economy. Deploying cloud infrastructure or broadband networks where the incumbent telco is a state security apparatus does not solve the pricing problem and may entrench it by strengthening the incumbent’s asset base without altering its pricing incentives. The soft-budget-constraint versus security-architecture diagnostic (Table 4) matters for FCS project design because it identifies where standard instruments are mismatched to the actual constraint, before funds are committed.

De-risking instruments and interconnection governance. The World Bank Group’s consolidated Guarantee Platform, launched in July 2024, provides political risk insurance, credit guarantees, and blended finance through MIGA and the IDA Private Sector Window [[World Bank Group, 2024b](#)]. These instruments lower the cost of capital for private operators investing in digital infrastructure, and they address a real barrier. But de-risking instruments that protect a fiber investment without conditioning on interconnection governance replicate the same necessary-but-not-sufficient problem that applies to cable capacity. A MIGA guarantee that lowers the cost of capital for a fiber operator in a market where the incumbent controls the landing station reduces investment risk without addressing the wholesale pricing bottleneck. The operational implication—that IFC equity positions and MIGA guarantees should be conditioned on verifiable interconnection governance milestones—is developed in Section 6 (Table 10).

Across these commitments, the pattern is consistent. The Bank’s digital infrastructure portfolio addresses real constraints at the physical access and demand layers. Wholesale interconnection

governance remains a gap in systematic programming. The analysis I develop here provides a diagnostic for identifying where that gap is binding.

7. Conclusion

Two years after the world's largest submarine cable system landed in Maputo, wholesale prices in Mozambique have not moved. The explanation is structural: the incumbent controls the landing station, so additional capacity strengthens its asset base without altering the terms on which domestic networks access it. The bargaining problem is unchanged. The prices are unchanged. Cable capacity is not the constraint. Competitive access to it is. The diagnostic error I aim to prevent is investing in bandwidth while leaving landing station governance and IXP aggregation scale unaddressed. Those are the variables the evidence identifies as operative.

Proposition 4 derives the scope conditions from the model's own structure: the aggregation mechanism operates if and only if networks can reach the Tier-1 carrier directly (C1), the market contains sufficient viable ISPs (C2), and the carrier optimizes over price rather than setting it administratively (C3). Where any condition fails, the model predicts no price response to aggregation investment, and the evidence is consistent with those predictions as it is with the mechanism where conditions hold. The null result in SOE-dominated markets and the positive excess in Uzbekistan are the model's predictions confirmed, not anomalies to explain away. A diagnostic that can distinguish these cases before investment is committed is what I contribute.

A direct statement of operational scope is warranted. The framework's strongest application is Sub-Saharan African markets with competitive gateway access—South Africa, Kenya, Rwanda, and the corridor markets where the 2Africa landing is reshaping commercial interconnection terms—and structural analogs elsewhere where Conditions (C0)–(C3) hold. In these markets, the diagnostic table (Table 9) applies directly. For SOE-dominated markets, the recommendation that structural reform precede infrastructure investment is directionally grounded in the Viet Nam and Pakistan case evidence and the soft-budget-constraint literature, but the operational pathway depends on whether the operative mechanism is rent-seeking gateway control or security-architecture pricing

(Table 4). For thin markets, the regional aggregation recommendation is theoretically correct but the minimum viable membership target must be recalculated at regional rather than national scale. Task teams operating outside the competitive African corridor should treat the model as a diagnostic lens identifying which constraint binds, not as a recipe whose parameters transfer directly.

The MOZIX prospective test—a pre-specified within-country comparison holding cable capacity constant while governance at the exchange point changes—will provide sharper evidence once the four-quarter evaluation window closes in late 2026. Exogenous variation in IXP establishment would allow causal identification of magnitude; such variation from regulatory mandates or phased rollouts is a priority for future research. More broadly, submarine cable landing stations, IXPs, and cloud on-ramps are increasingly co-located in carrier-neutral facilities. Based on current hyperscaler investment patterns, there is reason to expect that markets stuck at the Peering Cliff are also the markets least likely to attract the next wave of cloud on-ramp investment; if so, the digital infrastructure gap and the cloud access gap may be converging—a dynamic worth monitoring as hyperscaler deployment patterns become clearer.

The Peering Cliff is not primarily a technical problem. It is a bargaining problem with a structural solution.

Supplementary Appendices. Three appendices accompany this paper in the Supplementary Online Appendix:

- **Appendix A: Theoretical Derivation.** Full Nash bargaining setup, payoff functions, FOC derivation, four propositions with proofs, capacity extension, asymmetric bargaining power generalization, gateway lemma, aggregation extension with heterogeneous networks, and calibration exercise.
- **Appendix B: Case Studies and Portfolio Evidence.** Kenya and Nigeria long-run price trajectories, SOE case studies (Viet Nam, Pakistan), incumbent resistance cases, and World Bank portfolio analysis.
- **Appendix C: Replication Package.** Data sources, synthetic data generation procedure, and code documentation.

Data Availability Statement

TeleGeography IP transit pricing data are proprietary (subscription required). The analysis uses 22,503 city-level price observations from 126 countries (2010–2025), merged with IXP founding dates from Packet Clearing House, Internet Society, and national sources (149 countries, 1992–2024). SOE gateway classifications use the ITU DataHub Market Structure indicator (<https://datahub.itu.int>), verified against national regulatory authority filings. The replication package, available at the World Bank Microdata Library upon publication, includes: Python code for all analyses; the IXP founding dates database (fully public); the SOE classification table with ITU DataHub source codes; synthetic data matching summary statistics, distributional properties, and correlation structure of the TeleGeography panel; and instructions for obtaining TeleGeography data.

Reproducibility. The following outputs are fully reproducible without a TeleGeography subscription: Table 9; Table 4; Figures 5, 3, 2, 6 (conceptual diagrams or synthetic-data approximations); and

²⁷The Q4 2025 observation (\$3.50/Mbps) is plotted for completeness. A single post-expansion quarter is insufficient to distinguish a durable structural shift from transient volatility; this datum is preliminary and is not interpreted in the paper. Monitoring will continue through Q3 2026; see Section 5.1 and Supplementary Materials, Section S13 for the prospective design criteria.

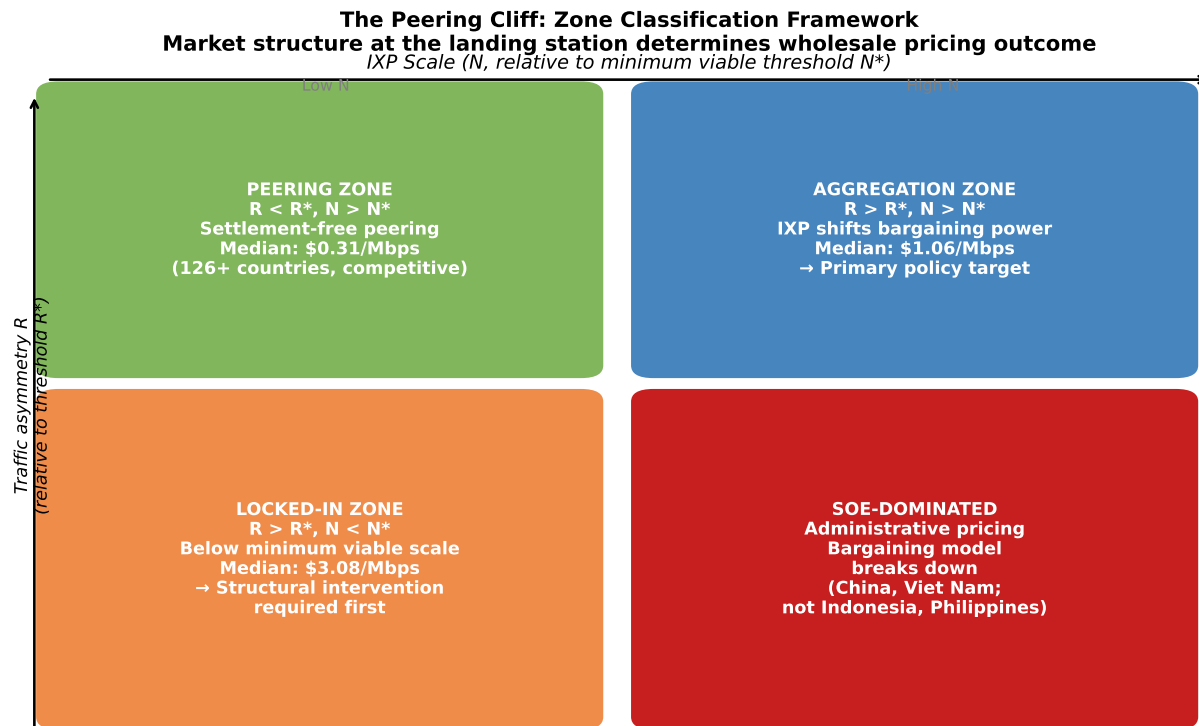


Figure 5: Capacity–Aggregation Complementarity: Framework Overview. Markets with high cable capacity but absent aggregation infrastructure (top-right quadrant) remain trapped at elevated wholesale prices; the policy lever is IXP investment, not additional cable capacity. The bottom-right quadrant represents the complementarity target: carrier-neutral aggregation combined with ample capacity. Country examples are illustrative. Generated by 01_figures_conceptual.py.

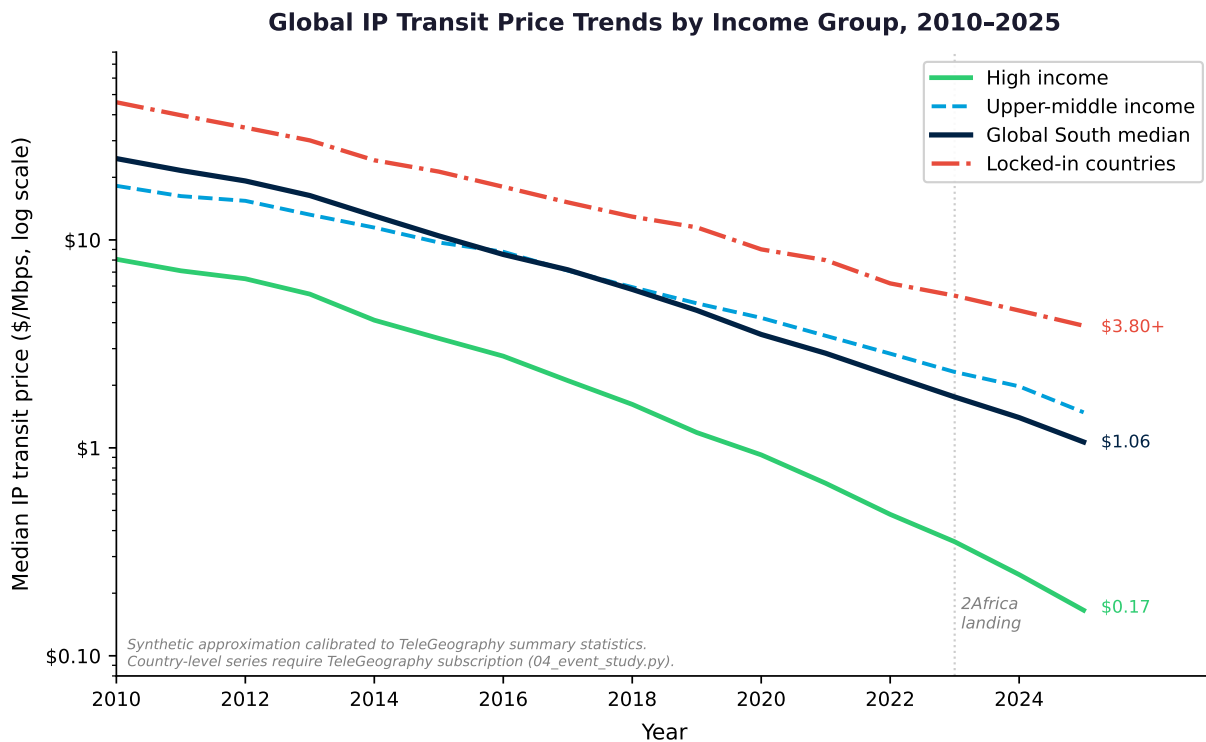


Figure 6: Global IP Transit Price Trends by Income Group, 2010–2025. Median prices (\$/Mbps, log scale) by World Bank income classification. High-income prices converge toward \$0.17/Mbps by 2025; the Global South median remains at \$1.06/Mbps; locked-in countries show persistent elevation above \$3/Mbps throughout the panel. *Note:* This figure is a synthetic approximation calibrated to TeleGeography summary statistics; country-level series require a TeleGeography subscription and are generated by 04_event_study.py.

Figure 3: Parallel Trends Assessment and Before--After Price Evidence (14-Country IXP Sample)

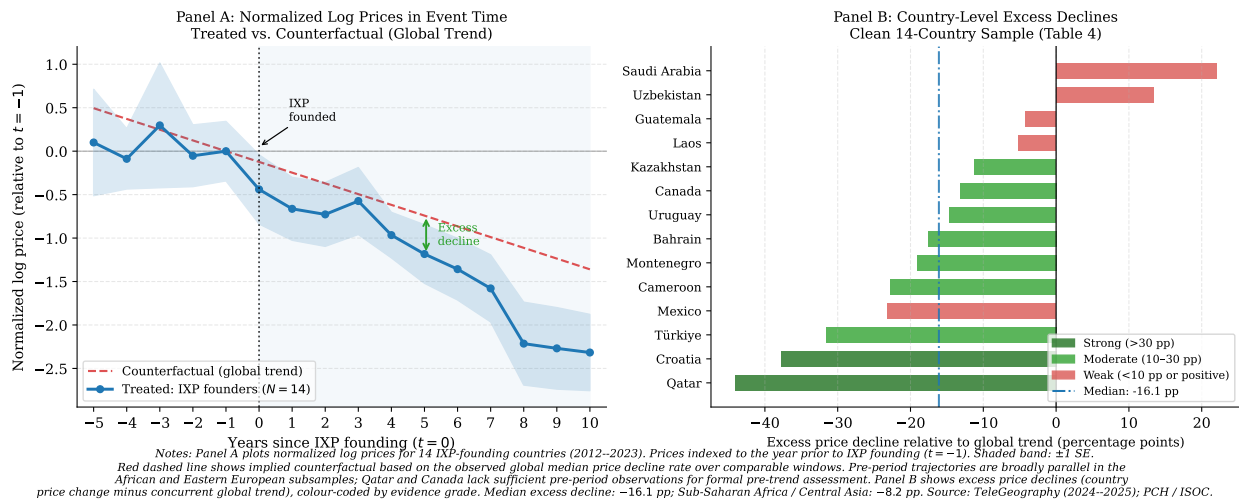


Figure 7: Parallel Trends Assessment and Before–After Price Evidence (14-Country IXP Sample). *Panel A:* Normalized log prices for 14 IXP-founding countries in event time, indexed to $t = -1$ (year prior to founding). Red dashed line: counterfactual trajectory based on the observed global median price decline rate. Shaded band: ± 1 SE. Pre-period trajectories are broadly parallel in the African and Eastern European subsamples; Qatar and Canada lack sufficient pre-period observations for formal assessment. *Panel B:* Country-level excess price declines (country price change minus concurrent global trend), colour-coded by evidence grade: Strong (>30 pp, dark green), Moderate (10–30 pp, light green), Weak (<10 pp or positive, red). Median excess: -16.1 pp (full 14-country sample); Sub-Saharan Africa / Central Asia primary estimate (Cameroon excluded): +1.1 pp; see Supplementary Materials, Table S5, footnote c. Source: TeleGeography IP Transit Pricing Service (2024–2025); IXP founding dates: Packet Clearing House / ISOC. Generated by 05_parallel_trends_figure.py.

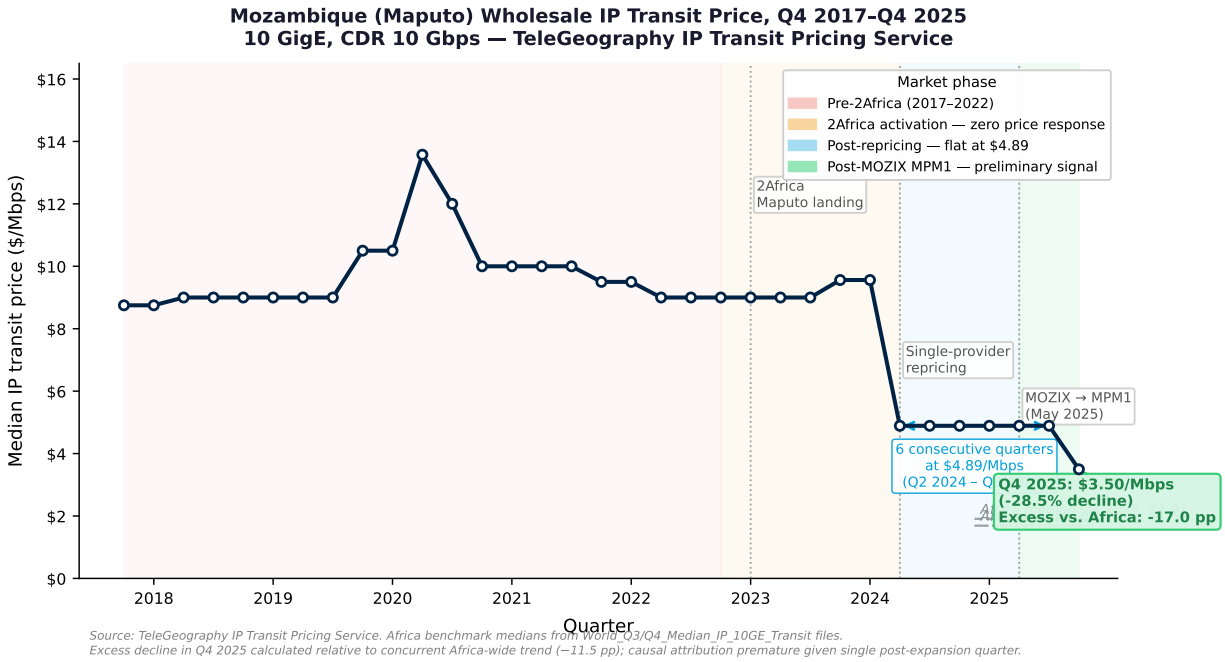


Figure 8: Mozambique (Maputo) Wholesale IP Transit Price, Q4 2017–Q4 2025. Median price per Mbps, 10 GigE circuit, CDR 10 Gbps. Three completed market phases are shaded: (i) pre-2Africa baseline (\$8.75–\$13.57/Mbps); (ii) 2Africa commercial activation (Q1 2023)—zero price response over six quarters; (iii) a single-provider repricing event in Q2 2024 establishing a new floor at \$4.89/Mbps, which held flat for six consecutive quarters. A fourth phase begins with MOZIX’s partial expansion to the carrier-neutral Digital Realty MPM1 facility (mid-2025).²⁷ *Interpretation:* The six-quarter flat at \$4.89/Mbps following 2Africa activation is the key datum for the paper’s central argument: abundant cable capacity produced no competitive pricing absent carrier-neutral aggregation. Source: TeleGeography IP Transit Pricing Service; Africa benchmarks from TeleGeography World Median IP Transit files. Generated by 06_mozambique_quarterly_analysis.py.

the Nash bargaining derivations in Supplementary Appendix A. Outputs requiring TeleGeography data: Tables 7, 8; Figure 8. The IXP founding dates database and governance classification are released without restriction.

Supplementary materials. Extended methodological details, SOE case studies, robustness checks, and the TTL Quick Reference are available at [URL to be added upon publication].

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